

Literature

- ML: Duda, Hart, Stork; Pattern Classification; Wiley Sons, 2001
- ML: C. M. Bishop; Neural Networks for Pattern Recognition, Oxford Univ. Press, 1995
- ML: T. M. Mitchell; Machine Learning, Mc Graw Hill, 1997
- ML: S. M. Kay; Fundamentals of Statistical Signal Processing, Prent. Hall, 1993
- Stat. Lern. Th.: V. N. Vapnik; Statistical Learning Theory, Wiley a. Sons, 1998
- Stat. Lern. Th.: Schölkopf, Smola; Learning with kernels, MIT Press, 2002
- Estimation Th.: S. M. Kay; Fundamentals of Statistical Signal Processing, Prentice Hall, 1993
- Estimation Th.: M. I. Jordan (ed.); Learning in Graphical Models, MIT Press, 1998

Notation

- Variable X, Y in uppercase letters are Random Variables.
- Vectors $\boldsymbol{v}$ in lowercase bold letters are column vectors:
$$\boldsymbol{v} = (v_1, v_2, \dots, v_n)^T$$
- Matrices $\boldsymbol{M}$ in uppercase bold letters consist of the set of vectors $\{\boldsymbol{m}\}$ such that $\boldsymbol{M} = \{\boldsymbol{m}\} = (\boldsymbol{m}_1, \dots, \boldsymbol{m}_n)^T$
 - i.e. a set of samples $\{\boldsymbol{x}\}$, where each $\boldsymbol{x}_i$ is a sample represented by some features, is written as $\boldsymbol{X}$, where each row is a sample.
- Estimates $\hat{\boldsymbol{v}}$ of true parameter vectors $\boldsymbol{v}$ are indicated with a hat.

Recap: Definition Expectation

- Expectation of a random variable X :
(discrete distribution, continuous distribution)

$$E(X) = \sum_j x_j p(x_j) \quad E(X) = \int_{-\infty}^{\infty} x p(x) dx \quad (1)$$

- Expectation of a function $g(X)$ with a random variable X :
(discrete distribution, continuous distribution)

$$E(g(X)) = \sum_j g(x_j) p(x_j) \quad E(g(X)) = \int_{-\infty}^{\infty} g(x) p(x) dx \quad (2)$$

- Variance of a random variable X :

$$\text{Var}(X) = E([X - E(X)]^2) = E(X^2) - E(X)^2 \quad (3)$$

Recap: Notation

- Be careful with notation
- Conditional Probability:

$$\begin{array}{ll} p(x; \omega) & x \text{ is a random variable, } \omega \text{ is a parameter} \\ p(x|y) & x \text{ is a random variable, } y \text{ is a random variable} \end{array} \quad (4)$$

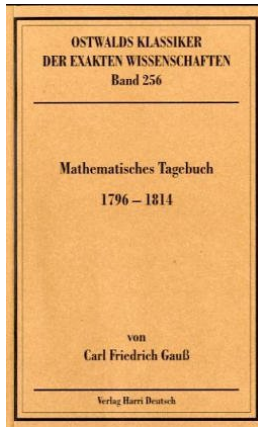
- Expectation (lots of different notations):

$$\begin{array}{ll} E, & E_X, & E_{p(x;\omega)}, & E_{p(x)\omega}, & E_{x \sim p(x;\omega)}, & E_{\omega}, & \dots \\ E_{\mathbf{X}}, & E_{p(\mathbf{x};\mathbf{w})}, & E_{\mathbf{x} \sim p(\mathbf{x};\mathbf{w})}, & E_{(\mathbf{x}_1, \mathbf{x}_2)}, & \dots \end{array} \quad (5)$$

- Notation you should use:

$$\begin{array}{ll} E, & E_{p(x;\omega)} \\ E_{\mathbf{X}}, & E_{p(\mathbf{x};\mathbf{w})} \end{array} \quad (6)$$

Carl Gauss (1777-1855)



Convergence in Deterministic Sequence

Convergence of a Deterministic Sequence

Let $a_1, a_2, \dots$ be a sequence of real numbers, and let a be another real number. We say that the sequence a_n converges to a , or $\lim_{n \rightarrow \infty} a_n = a$, if for every $\epsilon > 0$ there exists some n_0 such that

$$|a_n - a| \leq \epsilon, \quad \forall n \geq n_0.$$

Convergence in Probability

Definition

Let X_n be a sequence of random variables and let X be a random variable defined on a sample space. We say that X_n **converges in probability** to X if, for all $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P[|X_n - X| \geq \epsilon] = 0,$$

or equivalently,

$$\lim_{n \rightarrow \infty} P[|X_n - X| < \epsilon] = 1,$$

and write

$$X_n \xrightarrow{P} X.$$

Convergence in Distribution

Definition

Let X_n be a sequence of random variables, let X be a random variable. Let F_{X_n} and F_X be, respectively, the cdfs of X_n and X . Let $C(F_X)$ denote the set of all points where F_X is continuous. We say that X_n **converges in distribution** to X if

$$\lim_{n \rightarrow \infty} F_{X_n} = F_X(x) \quad \forall x \in C(F_X),$$

and write

$$X_n \xrightarrow{D} X.$$

Convergence with Probability 1

Convergence with Probability 1

Let $Y_1, Y_2, \dots$ be a sequence of random variables (not necessarily independent), and let Y be a random variable. We say that the sequence Y_n converges to Y **with probability 1** (or **almost surely**), if

$$P\left(\lim_{n \rightarrow \infty} Y_n = Y\right) = 1.$$

Strong Law of Large Numbers

Strong Law of Large Numbers

Let $X_1, X_2, \dots$ be a sequence of independent identically distributed random (i.i.d.) variables with mean μ . Then, the sequence of sample means $\bar{X}_n = (X_1 + \dots + X_n)/n$ converges to μ , **with probability 1**, in the sense that

$$P\left(\lim_{n \rightarrow \infty} \frac{X_1 + \dots + X_n}{n} = \mu\right) = 1.$$

Recap: Bias and Variance, Scalar Parameter

- Estimator:

$$\hat{w} = \hat{w}(\mathbf{X}) \quad (7)$$

- Evaluation criterion, mean squared error (MSE):

$$\text{mse}(\hat{w}, w) = \mathbb{E}_{\mathbf{X}}[(\hat{w} - w)^2] = \text{Var}(\hat{w}) + \text{Bias}^2(\hat{w}, w) \quad (8)$$

- **Variance:**

$$\text{Var}(\hat{w}) = \mathbb{E}_{\mathbf{X}}(\hat{w}^2) - \mathbb{E}_{\mathbf{X}}(\hat{w})^2 \quad (9)$$

- **Bias:**

$$\text{Bias}(\hat{w}, w) = \mathbb{E}_{\mathbf{X}}(\hat{w}) - w \quad (10)$$

- An estimator is **unbiased** if

$$\mathbb{E}_{\mathbf{X}}(\hat{w}) = w \quad (11)$$

i.e. on average over the training sets $\mathbf{X}$ the estimator will yield the true parameter.

Generalization Error

What does it mean to generalize?

- the **generalization error** which is the **expected loss on future data** should be as low as possible
- the generalization error (also called risk R) is the functional

$$R(g(\cdot; \mathbf{w})) = E_{\mathbf{z}}(L(y, g(\mathbf{x}; \mathbf{w}))) = \int_Z L(y, g(\mathbf{x}; \mathbf{w})) p(\mathbf{z}) d\mathbf{z}$$

with $p(\mathbf{z})$ the probability of $\mathbf{z} = (\mathbf{x}, y)$ and Z the set of all future $\mathbf{z}$.

Jensen's Inequality (1906)

If X is a random variable and h is a convex function, then

$$h(EX) \leq Eh(X).$$

SUR LES FONCTIONS CONVEXES ET LES INÉGALITÉS ENTRE
LES VALEURS MOYENNES¹

PAR

J. L. W. V. JENSEN
à COPENHAGUE.

1. *Des fonctions convexes et concaves. Définition. Exemples.*

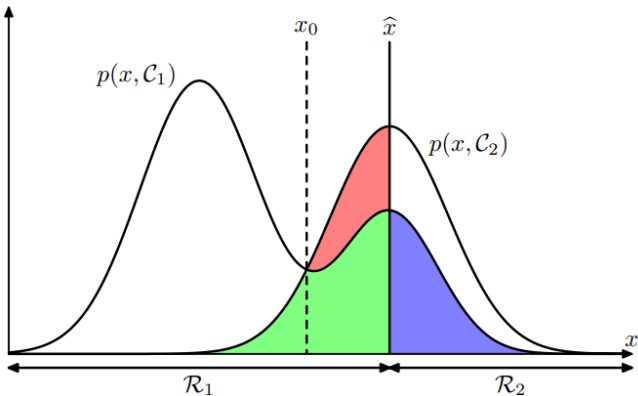
Dans sa célèbre Analyse algébrique (note II, pp. 457—59) CAUCHY démontre que «la moyenne géométrique entre plusieurs nombres est toujours inférieure à leur moyenne algébrique». La méthode employée par CAUCHY est extrêmement élégante, et elle a passé sans changement dans tous les traités d'analyse algébrique. Elle consiste, comme on sait, en ceci, que, de l'inégalité

$$\sqrt{ab} < \frac{1}{2}(a + b),$$

où a et b sont des nombres positifs, on est conduit à l'inégalité analogue pour quatre nombres, savoir

$$\sqrt[4]{abcd} < \frac{1}{4}(a + b + c + d),$$

Outlook - Decision Boundaries



Some more books...

Mathematics for Machine Learning

<https://mml-book.github.io/>

Understanding Machine Learning: From Theory to Algorithms

[https://www.cs.huji.ac.il/w~shais/
UnderstandingMachineLearning/](https://www.cs.huji.ac.il/w~shais/UnderstandingMachineLearning/)

Principles of ML Theory

<https://arxiv.org/abs/2106.10165>

Some mathematical statistics 0

Def: Random Variable

Consider a random experiment with a sample space C . A function X , which assigns to each element $c \in C$ one and only one number $X(c) = x$, is called a **random variable**.

Def: Random Sample

If the random variables $Y_1, Y_2, \dots, Y_n$ are independent and identically distributed (iid), then these random variables constitute a **random sample** of size n from the common distribution.

Source: Hogg, McKean & Graig: Intro. to Mathematical Statistics, 7th Edition

Some mathematical statistic I

Def: Statistic

Let $Y_1, Y_2, \dots, Y_n$ denote a sample on a random variable Y . Let $T = T(Y_1, Y_2, \dots, Y_n)$ be a function of the sample. Then T is called a **statistic**.

We now consider a random sample $Y_1, \dots, Y_n$ on a random variable Y with a density $p(y; \theta)$. We now consider a **point estimator** of θ . While we call T an estimator of θ , we call its realization t an **estimate** of θ .

Unbiasedness

Let $Y_1, Y_2, \dots, Y_n$ denote a sample on a random variable X with pdf $f(y; \theta)$, $\theta \in \Theta$. Let $T = T(Y_1, Y_2, \dots, Y_n)$ be a statistic. We say that T is an **unbiased estimator** of θ if $E(T) = \theta$.

Estimation Function - Estimate

- True parameter:

$$\theta \quad (12)$$

- Data:

$$X \sim p(\theta) \quad (13)$$

- Estimator:

$$\hat{\theta} \quad (14)$$

- Unbiased:

$$E_X(\hat{\theta}) = \theta \quad (15)$$

- Bias:

$$\text{bias}(\hat{\theta}) = E_X(\hat{\theta}) - \theta \quad (16)$$

- Variance:

$$\text{var}(\hat{\theta}) = E_X(\hat{\theta} - E_X(\hat{\theta}))^T (\hat{\theta} - E_X(\hat{\theta})) \quad (17)$$

A snapshot of the data generating process

- Some unknown process generates data $\mathbf{x} \in \mathbb{R}^d$.
- $\forall \mathbf{x}$ we want to infer some $y \in \mathbb{R}$.
- For some data $\{x_1, \dots, x_l\}$ we know the label $\{y_1, \dots, y_l\} \in \mathbb{R}$.
- Thus data $\mathbf{z} = (\mathbf{x}, y)$ are generated according to some joint **unknown** distribution $p(\mathbf{z}) = p(\mathbf{x}, y)$.
- If $\mathbf{z}$ is finite, $p(\mathbf{z})$ is the probability of observing $\mathbf{z}$.
- We call $\{\mathbf{z}_1, \dots, \mathbf{z}_l\}$ the training data.
- Training data are just a snapshot of the data.

Model Selection

- We assume there is a true, unknown labeling function
- We have a class of models.
- Select a model (function) that maps $y = g(\mathbf{x})$.
- For parametrized models we write $g(\mathbf{x}; \mathbf{w})$.
- We select the **best** model.
- The loss function measures the performance of our model.

$$L(y, g(\mathbf{x}; \mathbf{w})).$$

Generalization Error 1

- Error on one example only: **error, loss** ($L(y, g(\mathbf{x}; \mathbf{w}))$)
- Synonyms: **generalization error, true error, risk**
- The **generalization error** is the **expected loss** on future data

$$\begin{aligned} R(g(.; \mathbf{w})) &= \mathbb{E}_Z(L(y, g(\mathbf{x}; \mathbf{w}))) \\ &= \int_Z L(y, g(\mathbf{x}; \mathbf{w})) p(\mathbf{z}) d\mathbf{z} \\ &= \int_X \int_{\mathbb{R}} L(y, g(\mathbf{x}; \mathbf{w})) p(\mathbf{x}, y) dy d\mathbf{x} \end{aligned}$$

- $R(g(.; \mathbf{w}))$ is a functional.

Generalization Error 2

$R(g(\mathbf{x}; \mathbf{w}))$ is the expected loss for input $\mathbf{x}$:

$$\begin{aligned} R(g(\mathbf{x}; \mathbf{w})) &= \mathbb{E}_{y|\mathbf{x}}(L(y, g(\mathbf{x}; \mathbf{w}))) \\ &= \int_{\mathbb{R}} L(y, g(\mathbf{x}; \mathbf{w})) p(y|\mathbf{x}) dy \end{aligned}$$

Loss functions again...

- Quadratic Loss:

$$L(y, g(\mathbf{x}; \mathbf{w})) = (y - g(\mathbf{x}; \mathbf{w}))^2$$

- Zero-One Loss:

$$L(y, g(\mathbf{x}; \mathbf{w})) = \begin{cases} 0 & \text{for } y = g(\mathbf{x}; \mathbf{w}) \\ 1 & \text{for } y \neq g(\mathbf{x}; \mathbf{w}) \end{cases}$$

Generalization Error for a noisy function

- We assume y is a function of $\mathbf{x}$ perturbed by some noise:

$$y = f(\mathbf{x}) + \varepsilon$$

- Noise distribution $\varepsilon \sim p_{Noise}(\varepsilon)$
- $p(y|\mathbf{x}) = p_{Noise}(y - f(\mathbf{x}))$

We can compute the risk:

$$\begin{aligned} R(g(.; \mathbf{w})) &= \int_Z L(y, g(\mathbf{x}; \mathbf{w})) p(\mathbf{z}) d\mathbf{z} \\ &= \\ &= \dots \\ &= \int_X p(\mathbf{x}) \int_{\mathbb{R}} L(y, g(\mathbf{x}; \mathbf{w})) p_{Noise}(y - f(\mathbf{x})) dy d\mathbf{x} \end{aligned}$$

How can the generalization error be approximated?

- We don't have all future data.
- We can use a test set: $\{z_{l+1}, \dots, z_{l+m}\}$
- we assume the test set to be i.i.d. according to $p(\mathbf{x}, y)$

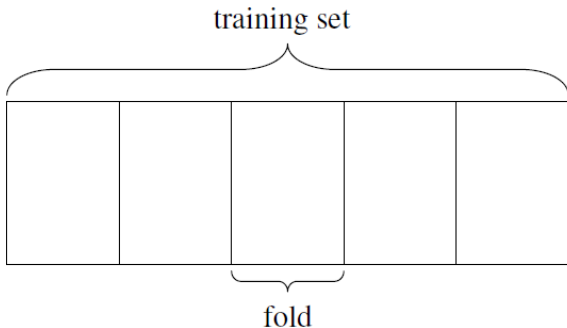
$$R_{emp}(g(.; \mathbf{w})) = \frac{1}{m} \sum_{i=l+1}^{l+m} L(y_i, g(\mathbf{x}_i; \mathbf{w}))$$

- $R_{emp}(g(.; \mathbf{w}))$ is an **estimate** for $R(g(.; \mathbf{w}))$
- By the **strong law of large numbers** (Appendix)
 $R_{emp}(g(.; \mathbf{w}))$ converges to $R(g(.; \mathbf{w}))$ for $m \rightarrow \infty$.

Cross validation

- We do not have enough data.
- n-fold cross validation
- LOO-CV Leave one out cross validation $n = l$
- **Cross validation is an almost unbiased estimator for the generalization error.**
- Proof: Luntz and Brailovsky, 1969

Cross validation

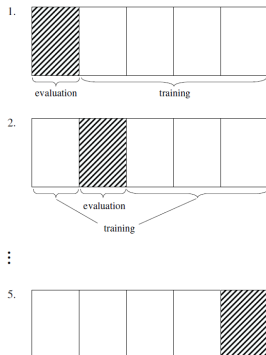


For the interested reader:

<https://www.jmlr.org/papers/volume5/grandvalet04a/grandvalet04a.pdf>

https://www.jstor.org/stable/2984809?seq=5#metadata_info_tab_contents

Cross validation



Bayes Rule, Bayes theorem, Bayes law

$$p(C_k|x) = \frac{p(x|C_k)p(C_k)}{p(x)}$$

$$posterior = \frac{likelihood \times prior}{evidence}$$

- posterior probability $p(C_k|x)$
- prior probability $p(C_k)$
- likelihood $p(x|C_k)$
- evidence or normalization factor

$$p(x) = p(x|C_1)P(C_1) + p(x|C_2)p(C_2)$$

- $P(C_1|x) + P(C_2|x) = 1$

Class conditional probability distribution

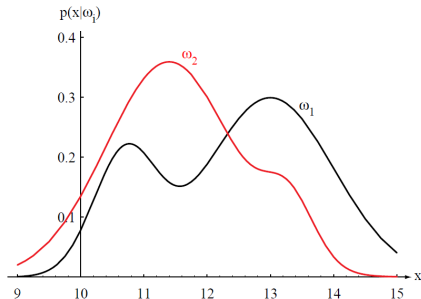


Figure 2.1: Hypothetical class-conditional probability density functions show the probability density of measuring a particular feature value x given the pattern is in category ω_i . If x represents the length of a fish, the two curves might describe the difference in length of populations of two types of fish. Density functions are normalized, and thus the area under each curve is 1.0.

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Bayes Decision Theory

- Two classes: w_1 and w_2
- Decision Rule: Decide for w_1 if $P(w_1) > P(w_2)$
- We observe a particular x and the probability of error is

$$P(error|x) = \begin{cases} P(w_1|x) & \text{if we choose } w_2 \\ P(w_2|x) & \text{if we choose } w_1 \end{cases}$$

- Average probability of error

$$P(error) = \int P(error|x) p(x) dx$$

- Bayesian decision rule for minimizing probability of error

decide w_1 if $P(w_1|x) > P(w_2|x)$; else decide w_2

Posterior probabilities

- Bayesian decision rule for minimizing probability of error

decide w_1 if $P(w_1|x) > P(w_2|x)$;
else decide w_2

- Equivalent Bayesian decision rule for minimizing probability of error

decide w_1 if $p(x|w_1)P(w_1) > p(x|w_2)P(w_2)$;
else decide w_2

- (about notation: here $P()$ is a probability, while $p()$ is a density....)

Class conditional probability distribution

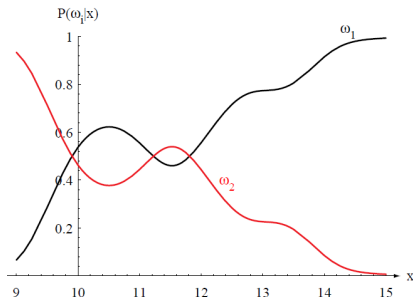
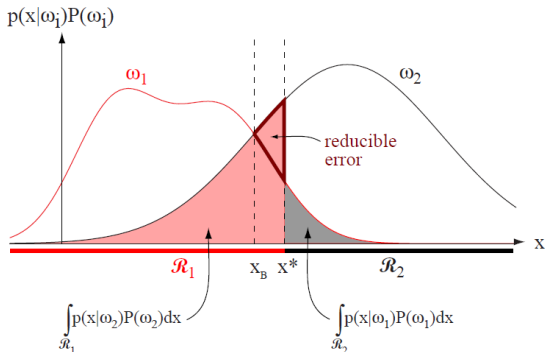


Figure 2.2: Posterior probabilities for the particular priors $P(\omega_1) = 2/3$ and $P(\omega_2) = 1/3$ for the class-conditional probability densities shown in Fig. 2.1. Thus in this case, given that a pattern is measured to have feature value $x = 14$, the probability it is in category ω_2 is roughly 0.08, and that it is in ω_1 is 0.92. At every x , the posteriors sum to 1.0.

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Components of Probability Error

$$P(\text{error}) = \int_{R_2} p(x|w_1) P(w_1) dx + \int_{R_1} p(x|w_2) P(w_2) dx$$



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Discrimination Functions

- Discrimination functions $g_i(\mathbf{x})$.
- A feature vector is being assigned to class C_i if

$$g_i(\mathbf{x}) > g_j(\mathbf{x}) \quad \forall j \neq i$$

- Decision regions $R_1, \dots, R_c$
- Decision boundaries separate regions
- Dichotomizer: define a single discriminant function

$$g(\mathbf{x}) \equiv g_1(\mathbf{x}) - g_2(\mathbf{x})$$

- Decide C_1 if $g(\mathbf{x}) > 0$

Minimal Risk for Gaussian Classification Task



Gauss again

Univariate

$$x \sim N(x; \mu, \sigma^2)$$

with density

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2}\right)$$

Multivariate

$$\mathbf{x} \sim N(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

with density

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{(1/2)}} \exp\left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

Gauss again - Try this!

$$E(x) = \int_{-\infty}^{\infty} N(x; \mu, \sigma^2) x dx = \dots$$

$$\text{var}(x) = \int_{-\infty}^{\infty} (x - \mu)^2 N(x; \mu, \sigma^2) dx = \dots$$

Hint 0: Try it out..

Hint 1: Changes of Variables..

Hint 2: Integration of odd functions is zero..

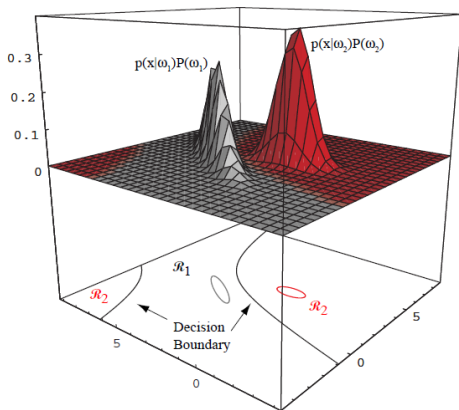
Hint 3: JKU Library ...

Minimal Risk for Gaussian Classification Task (MRGCT)

What it is about:

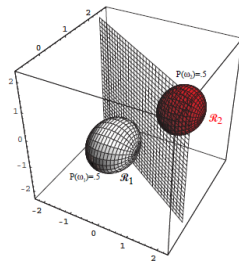
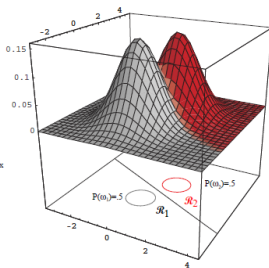
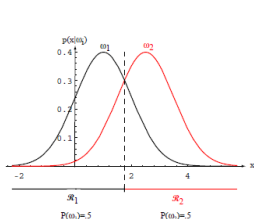
- Read: Duda, Hart, Stork; Pattern Classification; Wiley Sons, 2001
- We assume we know data is coming from two known Gaussian distributions.
- Binary classification task
- How do the discriminant functions look like?
- Calculate minimal risk.

Gaussian Classification Task



(Copyright © 2001 John Wiley Sons, Inc.)
(Notation $w_1 \equiv y = +1$ and $w_2 \equiv y = -1$)

Gaussian Classification Task



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MRGCT 1

- Class $y = +1$ data points

$$p(\mathbf{x}|y = +1) \propto N(\mu_{+1}, \Sigma_{+1})$$

- Class $y = -1$ data points

$$p(\mathbf{x}|y = -1) \propto N(\mu_{-1}, \Sigma_{-1})$$

- probability of observing a point from class $y = +1$ at $\mathbf{x}$

$$p(\mathbf{x}, y = +1)$$

- probability of observing a point from class $y = -1$ at $\mathbf{x}$

$$p(\mathbf{x}, y = -1)$$

- probability of observing a point at $\mathbf{x}$:

$$p(\mathbf{x}) = p(\mathbf{x}, y = +1) + p(\mathbf{x}, y = -1)$$

MRGCT 2

We define two regions

- of class +1 as $X_{+1} = \{\mathbf{x} | g(\mathbf{x}; \mathbf{w}) > 0\}$
- of class -1 as $X_{-1} = \{\mathbf{x} | g(\mathbf{x}; \mathbf{w}) < 0\}$

We use the zero-one loss

$$L(y, g(\mathbf{x}; \mathbf{w})) = \begin{cases} 0 & \text{for } y g(\mathbf{x}; \mathbf{w}) > 0 \\ 1 & \text{for } y g(\mathbf{x}; \mathbf{w}) < 0 \end{cases} \quad (18)$$

We obtain the risk:

$$\begin{aligned} R(g(\cdot; \mathbf{w})) &= \\ &= \int_{X_{+1}} p(\mathbf{x}, y = -1) d\mathbf{x} + \int_{X_{-1}} p(\mathbf{x}, y = +1) d\mathbf{x} \end{aligned}$$

MRGCT 3

$$\begin{aligned} R(g(., \mathbf{w})) &= \\ &= \int_{X_{+1}} p(y = -1, \mathbf{x}) d\mathbf{x} + \int_{X_{-1}} p(y = +1, \mathbf{x}) d\mathbf{x} \\ &= \int_{X_{+1}} p(y = -1|\mathbf{x}) p(\mathbf{x}) d\mathbf{x} + \int_{X_{-1}} p(y = +1|\mathbf{x}) p(\mathbf{x}) d\mathbf{x} \\ &= \int_X \left\{ \begin{array}{l} p(y = -1|\mathbf{x}) \text{ for } g(\mathbf{x}; \mathbf{w}) > 0 \\ p(y = +1|\mathbf{x}) \text{ for } g(\mathbf{x}; \mathbf{w}) < 0 \end{array} \right\} p(\mathbf{x}) d\mathbf{x} \end{aligned}$$

MRGCT 4

We find the optimal discriminant function:

$$g(\mathbf{x}, \mathbf{w}) = \begin{cases} > 0 & \text{for } p(y = +1|\mathbf{x}) > p(y = -1|\mathbf{x}) \\ < 0 & \text{for } p(y = -1|\mathbf{x}) > p(y = +1|\mathbf{x}) \end{cases} \quad (19)$$

The minimal risk is then:

$$R_{min} = \int_X \min\{p(y = -1|\mathbf{x}), p(y = +1|\mathbf{x})\} d\mathbf{x}$$

Optimal discrimination function

The optimal discriminant function is not unique

$$g(\mathbf{x}) = p(y = +1|\mathbf{x}) - p(y = -1|\mathbf{x})$$

or

$$g(\mathbf{x}) = \ln p(y = +1|\mathbf{x}) - \ln p(y = -1|\mathbf{x})$$

or:

$$g(\mathbf{x}) = \ln p(\mathbf{x}|y = +1) + \ln p(y = +1) - \ln p(\mathbf{x}|y = -1) - \ln p(y = -1)$$

Lets do some calculations

Now lets calculate this for Gaussians:

$$g(\mathbf{x}) = \ln p(\mathbf{x}|y = +1) + \ln p(y = +1) - \ln p(\mathbf{x}|y = -1) - \ln p(y = -1)$$

leads to

$$\begin{aligned} g(\mathbf{x}) &= \\ &+ \ln(N(\mu_{+1}, \Sigma_{+1})) + \ln p(y = +1) \\ &- \ln(N(\mu_{-1}, \Sigma_{-1})) - \ln p(y = -1) \\ &= \dots \\ &= \dots \\ &= -\frac{1}{2}\mathbf{x}^T A \mathbf{x} + \mathbf{w}^T \mathbf{x} + b \end{aligned}$$

Try this:

Minimal Risk for a Gaussian Classification Task



1 Introduction
2 Generalization Error
2.1 Model Quality
2.2 Gen. Error
2.2.1 Definition
2.2.2 Estimation
2.3 Minimal Risk
Example
3 Maximum Likelihood
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3.3 Cramer-Rao Lower Bound and Efficiency
3.4 Maximum Likelihood Estimation
3.5 Properties
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3.5.1 Invariant
3.5.2 MLE is Asymptotically Unbiased and Efficient
3.5.3 MLE is consistent for Zero CRLB
3.6 Expectation Maximization
3.7 Maximum Entropy Estimation

For Gaussians:

$$\begin{aligned}
 g(\mathbf{x}) &= -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_1)^T \Sigma_1^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) - \frac{d}{2} \ln 2\pi - \\
 &\quad \frac{1}{2} \ln |\Sigma_1| + \ln p(y=1) + \\
 &\quad \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_2)^T \Sigma_2^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) + \frac{d}{2} \ln 2\pi + \frac{1}{2} \ln |\Sigma_2| - \ln p(y=-1) = \\
 &\quad -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_1)^T \Sigma_1^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) - \frac{1}{2} \ln |\Sigma_1| + \ln p(y=1) + \\
 &\quad \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_2)^T \Sigma_2^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) + \frac{1}{2} \ln |\Sigma_2| - \ln p(y=-1) = \\
 &\quad -\frac{1}{2} \mathbf{x}^T \underbrace{(\Sigma_1^{-1} - \Sigma_2^{-1})}_A \mathbf{x} + \mathbf{x}^T \underbrace{(\Sigma_1^{-1} \boldsymbol{\mu}_1 - \Sigma_2^{-1} \boldsymbol{\mu}_2)}_w - \\
 &\quad \frac{1}{2} \boldsymbol{\mu}_1^T \Sigma_1^{-1} \boldsymbol{\mu}_1 + \frac{1}{2} \boldsymbol{\mu}_2^T \Sigma_2^{-1} \boldsymbol{\mu}_2 - \frac{1}{2} \ln |\Sigma_1| + \frac{1}{2} \ln |\Sigma_2| + \\
 &\quad \ln p(y=1) - \ln p(y=-1) = \\
 &\quad -\frac{1}{2} \mathbf{x}^T A \mathbf{x} + w^T \mathbf{x} + b
 \end{aligned}$$

Recap: Variance- Steiner'sche Verschiebungssatz

- Variance of a random variable X :

$$\text{Var}(X) = E([X - E(X)]^2) = E(X^2) - E(X)^2$$

- Try the proof yourself.

$$\text{Var}(x) = E([X - E(X)]^2) =$$

$$= E(X^2 - 2X E(X) + E(X)^2) =$$

$$= E(X^2) - 2E(X E(X)) + E(E(X)^2)$$

$$E(E(X)) = E(X)$$

$$= E(X^2) - 2E(X)^2 + E(X)^2 =$$

$$= E(X^2) - E(X)^2$$

Bias again

■ Bias:

$$\text{Bias}(\hat{w}, w) = E_X(\hat{w}) - w$$

■ Variance:

$$\text{var}(\hat{w}) = E_X \left((\hat{w} - E_X(\hat{w}))^T (\hat{w} - E_X(\hat{w})) \right) . \quad (20)$$

equivalent to :

$$\text{Var}(\hat{w}) = E_X(\hat{w}^2) - E_X(\hat{w})^2$$

■ MSE - Mean squared error:

$$\text{mse}(\hat{w}) = E_X \left((\hat{w} - w)^T (\hat{w} - w) \right) . \quad (21)$$

Even more on estimators

- **Unbiased estimator:**

$$E_X(\hat{w}) = w$$

- **Asymptotically unbiased estimator** for training set with size l

$$E_X(\hat{w}) \xrightarrow{l \rightarrow \infty} w$$

- **Consistent estimator:** for large training sets the estimator approaches the true value.

$$\hat{w} \xrightarrow{l \rightarrow \infty} w$$

Even more on estimators

- **Efficient estimator:**

$$\text{covar}(\hat{\mathbf{w}}) = \text{CRLB} .$$

- **Asymptotically efficient estimator:**

$$\text{covar}(\hat{\mathbf{w}}) \xrightarrow{l \rightarrow \infty} \text{CRLB} .$$

Consistent Estimator

Definition

Let X be a random variable with cdf $F(x, \theta)$, $\theta \in \Theta$. Let $X_1, \dots, X_n$ be a sample from the distribution of X and let T_n denote a statistic. We say T_n is a **consistent** estimator of θ if T_n converges in probability to θ

$$T_n \xrightarrow{P} \theta.$$

Reminder

$$\lim_{n \rightarrow \infty} P[|T_n - \theta| \geq \epsilon] = 0.$$

Bias-Variance Decomposition I

$$\begin{aligned}\text{mse}(\hat{\boldsymbol{w}}) &= \mathbb{E}_{\boldsymbol{X}} \left((\hat{\boldsymbol{w}} - \boldsymbol{w})^T (\hat{\boldsymbol{w}} - \boldsymbol{w}) \right) \\ &= \dots \\ &\quad \text{var}(\hat{\boldsymbol{w}}) + b^2(\hat{\boldsymbol{w}}) .\end{aligned}$$

(Source: Neural Networks and the Bias Variance Dilemma by
Geman et al. 1992 paper) <https://www.dam.brown.edu/people/documents/bias-variance.pdf>

Bias-Variance Decomposition II

$$\begin{aligned} \text{mse}(\hat{\mathbf{w}}) &= \mathbb{E}_{\mathbf{X}} \left((\hat{\mathbf{w}} - \mathbf{w})^T (\hat{\mathbf{w}} - \mathbf{w}) \right) = \\ &\mathbb{E}_{\mathbf{X}} \left(((\hat{\mathbf{w}} - \mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}})) + (\mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}}) - \mathbf{w}))^T \right. \\ &\quad \left. ((\hat{\mathbf{w}} - \mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}})) + (\mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}}) - \mathbf{w})) \right) = \\ &\mathbb{E}_{\mathbf{X}} \left((\hat{\mathbf{w}} - \mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}}))^T (\hat{\mathbf{w}} - \mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}})) - \right. \\ &\quad 2 (\hat{\mathbf{w}} - \mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}}))^T (\mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}}) - \mathbf{w}) + \\ &\quad \left. (\mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}}) - \mathbf{w})^T (\mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}}) - \mathbf{w}) \right) = \\ &\mathbb{E}_{\mathbf{X}} \left((\hat{\mathbf{w}} - \mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}}))^T (\hat{\mathbf{w}} - \mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}})) \right) + \\ &\quad (\mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}}) - \mathbf{w})^T (\mathbb{E}_{\mathbf{X}}(\hat{\mathbf{w}}) - \mathbf{w}) = \\ &\text{var}(\hat{\mathbf{w}}) + b^2(\hat{\mathbf{w}}) . \end{aligned} \tag{22}$$

Bias - Variance Decomposition

true parameter w
estimator $\hat{w}$

1. $\pm E(\hat{w})$ (aktive Null)
2. Binomische Formel
3. nur $\hat{w}$ von x abhängig,
Rest irrelevant für E

$$mse(\hat{w}) = E((\hat{w} - w)^T (\hat{w} - w)) =$$

$$= E((\hat{w} - E(\hat{w})) + (E(\hat{w}) - w)^T ((\hat{w} - E(\hat{w})) + (E(\hat{w}) - w)))$$

$$= E((\hat{w} - E(\hat{w}))^2 - 2(\hat{w} - E(\hat{w}))^T \cdot (E(\hat{w}) - w)$$

$$+ (E(\hat{w}) - w)^2) =$$

$$= \underbrace{E((\hat{w} - E(\hat{w}))^2)}_{\text{var}(\hat{w})} - E(2(\hat{w} - E(\hat{w}))^T \cdot (E(\hat{w}) - w)) + \underbrace{E((E(\hat{w}) - w)^2)}_{\text{bias}(\hat{w})^2} = \text{var} + \text{bias}^2$$

$$E(2(\hat{w} - E(\hat{w}))^T (E(\hat{w}) - w)) =$$

$$= 2 E((\hat{w} - E(\hat{w}))^T (E(\hat{w}) - w)) =$$

$$= 2 (\underbrace{E(\hat{w}) - E(\hat{w})}_0)^T (E(\hat{w}) - w) = 0$$

Minimal Variance Unbiased Estimator MVUE

- We want to find an optimal estimator - an unbiased estimator with minimal variance: MVUE.
- An estimator is said to be **efficient** if it reaches the **Cramer-Rao Lower Bound CRLB**.
- Not every MVUE is efficient.

Score Function

For a parametric model $p(x; \theta)$ with parameter θ we define the **score function** as

$$z(x, \theta) \equiv \frac{\partial}{\partial \theta} \ln p(x; \theta).$$

with

$$z'(x, \theta) \equiv \frac{\partial^2}{\partial \theta^2} \ln p(x; \theta).$$

and we can show that

$$\mathbb{E}_\theta(z(x, \theta)) = 0,$$

and

$$\text{VAR}(z(x, \theta)) = -\mathbb{E}(z'(x, \theta)).$$

Score Function - Proof 1

Proof:

$$z(x, \theta)p(x; \theta) = \left(\frac{\partial}{\partial \theta} \ln p(x; \theta) \right) p(x; \theta) = \frac{\frac{\partial}{\partial \theta} p(x; \theta)}{p(x; \theta)} p(x; \theta) = \frac{\partial}{\partial \theta} p(x; \theta).$$

Then, since $\int p(x; \theta) dx = 1$

$$\mathbb{E}_{\theta}(z(X, \theta)) = \int z(x, \theta)p(x; \theta)dx = \int \frac{\partial}{\partial \theta} p(x; \theta)dx = \frac{\partial}{\partial \theta} \int p(x; \theta)dx = 0.$$

Score Function - Proof 2

$$\begin{aligned} 0 &= \frac{\partial}{\partial \theta} \mathbb{E}[z(X, \theta)] \\ &= \frac{\partial}{\partial \theta} \int z(x, \theta) p(x; \theta) dx \quad \text{, } z(x, \theta) p(x; \theta) \text{ (see proof 1)} \\ &= \int \left(z'(x, \theta) p(x; \theta) + z(x, \theta) \left(\frac{\partial}{\partial \theta} p(x; \theta) \right) \right) dx \\ &= \int (z'(x, \theta) p(x; \theta) + z(x, \theta)^2 p(x; \theta)) dx \\ &= \mathbb{E}[z'(X, \theta)] + \mathbb{E}[z(X, \theta)^2] \\ &= \mathbb{E}[z'(X, \theta)] + \text{VAR}[z(X, \theta)] \end{aligned}$$

Fisher Information I

The Fisher information gives the amount of information that an observable random variable x carries about an unobservable parameter w upon which the parameterized density function $p(x; w)$ of x depends.

Fisher Information II

The Fisher information matrix $\mathbf{I}_F$ for a parameterized model is

$$\mathbf{I}_F(\mathbf{w}) : [\mathbf{I}_F(\mathbf{w})]_{ij} = \mathbb{E}_{p(\mathbf{x}; \mathbf{w})} \left(\frac{\partial \ln p(\mathbf{x}; \mathbf{w})}{\partial w_i} \frac{\partial \ln p(\mathbf{x}; \mathbf{w})}{\partial w_j} \right), \quad (23)$$

Fisher Information III

If the density function $p(\mathbf{x}; \mathbf{w})$ satisfies

$$\forall \mathbf{w} : \mathbb{E}_{p(\mathbf{x}; \mathbf{w})} \left(\frac{\partial \ln p(\mathbf{x}; \mathbf{w})}{\partial \mathbf{w}} \right) = \mathbf{0} \quad (24)$$

then the Fisher information matrix is

$$\mathbf{I}_F(\mathbf{w}) : \mathbf{I}_F(\mathbf{w}) = - \mathbb{E}_{p(\mathbf{x}; \mathbf{w})} \left(\frac{\partial^2 \ln p(\mathbf{x}; \mathbf{w})}{\partial \mathbf{w} \partial \mathbf{w}} \right) . \quad (25)$$

Cramer-Rao Lower Bound (CRLB)

Assume that

$$\forall \mathbf{w} : \mathbb{E}_{p(\mathbf{x}; \mathbf{w})} \left(\frac{\partial \ln p(\mathbf{x}; \mathbf{w})}{\partial \mathbf{w}} \right) = \mathbf{0} \quad (26)$$

and that the estimator $\hat{\mathbf{w}}$ is unbiased.

Then,

$$\text{covar}(\hat{\mathbf{w}}) - \mathbf{I}_F^{-1}(\mathbf{w}) \quad (27)$$

is positive definite:

$$\text{covar}(\hat{\mathbf{w}}) - \mathbf{I}_F^{-1}(\mathbf{w}) \geq \mathbf{0} . \quad (28)$$

Proof of the CRLB I

- Proof Idea: use Cauchy-Schwarz Inequality and derivation of score function.
- $X_1, \dots, X_n$ iid
- T is an unbiased estimator of θ based on $X_1, \dots, X_n$
- Idea: Cauchy-Schwarz Inequality:

$$[\text{covar}(Z, Y)]^2 \leq (\text{VAR}(Z))(\text{VAR}(Y)) \quad (29)$$

- Choose Y to be estimator T of θ
- Choose Z to be quantity $Z \equiv \sum z(X_i, \theta) = \sum \frac{\partial}{\partial \theta} \ln p(x; \theta)$

Proof of the CRLB II

- $E(Z) = 0$ and $\text{VAR}(Z) = nI(\theta)$

$$E[T] = \int \cdots \int T(x_1, \dots, x_n) p(x_1; \theta) \cdots p(x_n; \theta) dx_1 \cdots dx_n$$

- differentiate the mean with respect to θ

$$\begin{aligned} 1 &= \int \cdots \int T(x_1, \dots, x_n) \left[\sum \frac{\partial p(x_i; \theta)}{\partial \theta} \right] \\ &\quad p(x_1; \theta) \cdots p(x_n; \theta) dx_1 \cdots dx_n \\ &= E[TZ] \end{aligned}$$

- $E[TZ] \leq nI(\theta)\text{VAR}(Y)$ **qed.**

Frame Title

- Theorem: For any two random variables X and Y it can be shown that $\text{covar}(X, Y) = E(XY) - \mu_X \mu_Y$.

$$\begin{aligned}\text{covar}(X, Y) &= E((X - E(X))(Y - E(Y))) = \\&= E(XY - XE(Y) - E(X)Y + E(X)E(Y)) = \\&= E(XY) - E(XE(Y)) - E(E(X)Y) + E(E(X)E(Y)) = \\&= E(XY) - 2E(X)E(Y) + E(X)E(Y) = \\&= E(XY) - E(X)E(Y) = E(XY) - \mu_X \mu_Y\end{aligned}$$

Maximum Likelihood Principle

- Dataset of l independent observations $\{\mathbf{x}\} = \mathbf{x}_1, \dots, \mathbf{x}_l$
- Likelihood of the dataset $\mathcal{L}(\{\mathbf{x}\}; \mathbf{w}) = p(\mathbf{x}; \mathbf{w})$ is the probability of the model $p(\mathbf{x}; \mathbf{w})$ to produce the data
- For iid data

$$\mathcal{L}(\{\mathbf{x}\}; \mathbf{w}) = p(\{\mathbf{x}\}; \mathbf{w}) = \prod_{i=1}^l p(\mathbf{x}_i; \mathbf{w})$$

- $\mathbf{w}_{ML}$ maximizes the likelihood function

$$\mathbf{w}_{ML} = \operatorname{argmax} \quad \mathcal{L}(\{\mathbf{x}\}; \mathbf{w})$$

- Negative log likelihood

$$-\ln \mathcal{L}(\{\mathbf{x}\}; \mathbf{w}) = -\ln \prod_{i=1}^l p(\mathbf{x}_i; \mathbf{w}) = -\sum_{i=1}^l \ln p(\mathbf{x}_i; \mathbf{w})$$

MLE for Gaussian Distributions

- Assume iid data $x_1, \dots, x_l$ from $N(\mu, \sigma^2)$.
- Write down the log likelihood function $\ln \mathcal{L}(\{x\}; \mu, \sigma)$.
- How does the ML solution for μ_{ML} look like?
- How does the ML solution for σ_{ML}^2 look like?
- Derive $E(\mu_{ML})$.
- Derive $E(\sigma_{ML}^2)$.
- What are the *sufficient* statistics for the Gaussian distribution?

MLE for Gaussian $\rightarrow$ won't be relevant

$$f(\mu) = \left(\sum \ln \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}} \right) \right)$$

$$\frac{\partial f}{\partial \mu} = \frac{\partial}{\partial \mu} \left(n \ln \frac{1}{\sqrt{2\pi\sigma^2}} + \sum_{i=1}^n \left(-\frac{(x_i - \mu)^2}{2\sigma^2} \right) \right) =$$

$$= \frac{\partial}{\partial \mu} \left(-\frac{1}{2\sigma^2} \cdot \sum_{i=1}^n (x_i - \mu)^2 \right) =$$

$$= -\frac{1}{2\sigma^2} \cdot \sum_{i=1}^n \frac{\partial}{\partial \mu} (x_i - \mu)^2 =$$

$$= -\frac{1}{2\sigma^2} \sum_{i=1}^n -1 \cdot 2 \cdot (x_i - \mu) =$$

$$= -\frac{1}{2\sigma^2} \cdot \sum_{i=1}^n 2(\mu - x_i) =$$

$$= -\frac{1}{\sigma^2} \cdot \sum_{i=1}^n (\mu - x_i) =$$

$$= -\frac{1}{\sigma^2} \cdot \left(n \cdot \mu - \sum_{i=1}^n x_i \right) \stackrel{!}{=} 0$$

$$-\frac{1}{\sigma^2} \cdot n \cdot \mu + \frac{1}{\sigma^2} \sum_{i=1}^n x_i = 0$$

$$-\cancel{\frac{1}{\sigma^2}} n \cdot \mu = -\cancel{\frac{1}{\sigma^2}} \sum_{i=1}^n x_i$$

$$\mu = \frac{\sum_{i=1}^n x_i}{n}$$

$$f(\mu, \sigma) = \left(\sum \ln \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}} \right) \right)$$

$$\frac{\partial f}{\partial \sigma^2} \stackrel{?}{=} \frac{\partial}{\partial \sigma^2} \left(\sum_{i=1}^n \left(\ln \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) + \ln \left(e^{-\frac{(x_i - \mu)^2}{2\sigma^2}} \right) \right) \right) =$$

$$= - \left(\sum_{i=1}^n \frac{\partial}{\partial \sigma^2} \left(\ln \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) \right) + \frac{\partial}{\partial \sigma^2} \left(\ln \left(e^{-\frac{(x_i - \mu)^2}{2\sigma^2}} \right) \right) \right) =$$

$$= - \left(\sum_{i=1}^n \left(\frac{\partial}{\partial \sigma^2} \left(\ln \left(\frac{1}{\sqrt{2\pi}} \right) + \ln \left(\frac{1}{\sigma^2} \right) \right) + \frac{\partial}{\partial \sigma^2} \left(- \frac{(x_i - \mu)^2}{2\sigma^2} \right) \right) \right) =$$

$$= - \sum_{i=1}^n \left(\frac{\partial}{\partial \sigma^2} \ln \left(\frac{1}{\sigma^2} \right) + \left(- \frac{(x_i - \mu)^2}{2} \cdot \frac{\partial}{\partial \sigma^2} \left(\frac{1}{\sigma^2} \right) \right) \right) =$$

$$= - \sum_{i=1}^n \left(- \frac{1}{\sigma^2} + \left(- \frac{(x_i - \mu)^2}{2} \cdot - \frac{1}{(\sigma^2)^2} \right) \right) =$$

$$= - \sum_{i=1}^n \left(- \frac{1}{\sigma^2} + \frac{(x_i - \mu)^2}{2\sigma^4} \right) =$$

$$= - \sum_{i=1}^n \frac{-2\sigma^2 + (x_i - \mu)^2}{2\sigma^4} =$$

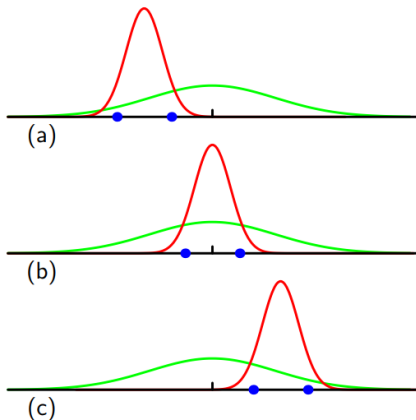
$$= \sum_{i=1}^n \frac{2\sigma^2 - (x_i - \mu)^2}{2\sigma^4} \stackrel{!}{=} 0$$

$$\frac{1}{2\sigma^4} \cdot \sum_{i=1}^n (2\sigma^2 - (x_i - \mu)^2) \stackrel{!}{=} 0$$

$$\frac{1}{2\sigma^4} \cdot \left(n \cdot 2\sigma^2 - \sum_{i=1}^n (x_i - \mu)^2 \right) \stackrel{!}{=} 0$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{2n}$$

Bias in ML estimator



(Copyright @ Bishop 1.15)

MLE for Poisson Distribution

- Assume you have n samples from a Poisson distribution $X_i \sim \text{Pois}(\lambda)$.
- How does the log-likelihood function look like?
- What is the MLE for λ ?
- Derive the Fischer Information for the Poisson distribution?



ML for Poisson Distribution

$$f(\lambda) = - \sum_{i=1}^n \ln \left(\frac{\lambda^{k_i} e^{-\lambda}}{k_i!} \right) = - \sum_{i=1}^n \ln(\lambda^{k_i}) + \ln(e^{-\lambda}) - \ln(k_i!) =$$

$$= - \left(\sum_{i=1}^n (k_i \cdot \ln(\lambda)) \right) + n \cdot -\lambda - \sum_{i=1}^n \ln(k_i!)$$

$$\frac{\partial}{\partial \lambda} f(\lambda) = - \left(\left(\sum_{i=1}^n k_i \frac{\partial}{\partial \lambda} \ln(\lambda) \right) + n \cdot \frac{\partial}{\partial \lambda} \lambda \right) =$$

$$= - \left(\left(\sum_{i=1}^n k_i \cdot \frac{1}{\lambda} \right) + n \right) \stackrel{!}{=} 0$$

$$\frac{1}{\lambda} \cdot \left(\sum_{i=1}^n k_i \right) = +n$$

$$\frac{1}{\lambda} = + \frac{n}{\sum_{i=1}^n k_i}$$

$$\lambda = + \frac{\sum_{i=1}^n k_i}{n}$$

Pros and Cons for MLE

Cons

- MLE does not need to exist.
- MLE does not need to be unique.

Pros

- MLE is **invariant** under parameter change.
- MLE is **asymptotically unbiased** and **asymptotically efficient**.
- MLE is **consistent** if CRLB is zero.

Invariance Property of MLEs

Theorem

If $\hat{\theta}$ is the MLE of θ , then for any function $\tau(\theta)$, the MLE of $\tau(\theta)$ is $\tau(\hat{\theta})$.

MLE Asymptotic Properties

Theorem

If $p(\boldsymbol{x}; \boldsymbol{w})$ satisfies

$$\forall \boldsymbol{w} : \mathbb{E}_{p(\boldsymbol{x}; \boldsymbol{w})} \left(\frac{\partial \ln p(\boldsymbol{x}; \boldsymbol{w})}{\partial \boldsymbol{w}} d\boldsymbol{x} \right) = \mathbf{0} \quad (30)$$

then the MLE which maximizes $p(\{\boldsymbol{x}\}; \boldsymbol{w})$ is asymptotically distributed according to

$$\hat{\boldsymbol{w}} \stackrel{l \rightarrow \infty}{\propto} \mathcal{N}(\boldsymbol{w}, \boldsymbol{I}_F^{-1}(\boldsymbol{w})) , \quad (31)$$

where $\boldsymbol{I}_F(\boldsymbol{w})$ is the Fisher information matrix evaluated at the unknown parameter $\boldsymbol{w}$.

Some ideas for exam questions

- What does every probability density function need to fulfill?
- State Jensen's inequality.
- What is the CRLB and why is it useful?
- State the Cauchy-Schwarz Inequality?
- MLE for Poisson distribution
- MLE for Gaussian distribution
- Describe the score function and its properties.
- Derive the bias-variance decomposition.
- How is risk defined?
- What are the properties of the MLE?

CAUCHY SCHWARZ INEQUALITY

$$|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$$

Literature

- T. M. Cover, J. A. Thomas, Elements of Information Theory, 2006.
- E. T. Jaynes, Information Theory and Statistical Mechanics, 1957
- C. E. Shannon, A Mathematical Theory of Communication, 1948
- S. Kullback, R.A. Leibler, On information and Sufficiency, 1951

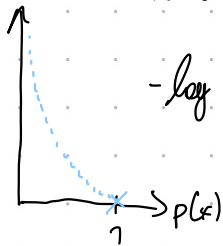
Information Content

What is the information content of event x ?

(Please follow the drawing on the black board)

Event x
self information

($\neq$ mutual information)



$-\log p(x)$

$$p(x) = 1$$

} no information
no insecurity
small $p(x) \rightarrow$ more information
learn a lot $\leftarrow$

x_1, x_2 independent $\rightarrow$ Information
should sum up

Entropy - Definition

For discrete random variables $p_k = p(x = x_k)$:

$$H(X) = - \sum_{k \geq 1} p_k \log p_k . \quad (32)$$

We assume $p_k \log p_k = 0$ for $p_k = 0$.

For continuous random variables x with probability density $p(x)$:

$$H(X) = - \int_{-\infty}^{\infty} p(x) \log p(x) dx , \quad (33)$$

where we set $p(x) \log p(x) = 0$ for $p(x) = 0$.

(About notation: $H(X)$ or $H(p)$)

Uniform distribution

The discrete distribution with maximum entropy is the uniform distribution.

$$H(X) = \dots$$

(Please fill this out yourself or follow the writing on the blackboard.)

$$p_k = \frac{1}{k}$$

$$H(X) = - \sum_{k=1}^{\infty} p_k \log p_k =$$

$$= - \sum_{k=1}^{\infty} \frac{1}{k} \log \left(\frac{1}{k} \right) = - \log \left(\frac{1}{\infty} \right) =$$

$$= \log(\infty)$$

Binary entropy function

Given a binary random variable, derive the entropy.

$$H(X) =$$

(Please fill this out yourself or follow the writing on the blackboard.)

$$p_1 = p(X=1) = \theta$$

$$p_0 = p(X=0) = 1 - \theta$$

$$H(X) = - (p_1 \log p_1 + p_0 \log p_0) =$$

$$= - (\theta \log \theta + (1-\theta) \log (1-\theta))$$

Cross-Entropy - Definition

The *cross entropy* between two distributions p_1 and p_2 is defined as

$$H^*(p_1, p_2) = - \int_U p_1(\mathbf{u}) \ln p_2(\mathbf{u}) d\mathbf{u} . \quad (34)$$

Joint Entropy

For two discrete random variables X and Y the joint entropy is defined as:

$$H(X, Y) = - \sum_{x,y} p(x, y) \log p(x, y) . \quad (35)$$

Kullback-Leibler divergence

KL divergence, information gain, relative entropy $KL(p_1, p_2)$ is defined as

$$D_{KL}(p_1 \parallel p_2) = \int_U p_1(\mathbf{u}) \ln \frac{p_1(\mathbf{u})}{p_2(\mathbf{u})} d\mathbf{u} \quad (36)$$

$$D_{KL}(p_1 \parallel p_2) = \sum_{k=1}^K p_1(\mathbf{u}_k) \ln \frac{p_1(\mathbf{u}_k)}{p_2(\mathbf{u}_k)} \quad (37)$$

(What is the difference between a divergence measure and a metric? What is Jeffreys' divergence?)

METRIC

$d: (X, X) \rightarrow \mathbb{R}$ metric $(\Rightarrow)$

(1) $d(x, y) = 0 \Leftrightarrow x = y$

(2) $d(x, y) = d(y, x)$

(3) $d(x, z) \leq d(x, y) + d(y, z)$

$\Rightarrow D_{KL}$ not a metric because (2), (3) are not fulfilled

Kullback-Leibler divergence II

- KL divergence is always greater than or equal to zero:
PROOF using Jensen's Inequality.

$$D_{\text{KL}}(p_1 \parallel p_2) \geq 0 \quad (38)$$

- KL divergence can be written as sum of negative entropy and the cross entropy

$$D_{\text{KL}}(p_1 \parallel p_2) = -H(p_1) + H^*(p_1, p_2) \quad (39)$$

<https://projecteuclid.org/journals/annals-of-mathematical-statistics/volume-22/issue-1/On-Information-and-Sufficiency/10.1214/aoms/1177729694.full>

PROOF OF 38

$$-D_{KL} = - \int_X \log \left(\frac{p_2(x)}{p_1(x)} \right) p_1(x) dx =$$

$$= \int_X \log \frac{p_2(x)}{p_1(x)} p_1(x) dx$$

Sensen
↓

$$\leq \log \int_X \frac{p_2(x)}{p_1(x)} p_1(x) dx =$$

$$= \log \int_X p_2(x) dx = \log 1 = 0$$

Mutual Information

How dependent are two random variables X and Y ?

$$I(X; Y) = D_{\text{KL}}(p(x, y) \parallel p(x)p(y))$$

- $I(X; Y) \geq 0$
- KL divergence between joint and factored marginal distribution
- MI measures information gain if we update from a model that treats two variables independent to one that models their joint density
- $I(X; Y) = H(X) + H(Y) - H(X, Y)$

Principle of Maximum Entropy - Jaynes

"The quantity x is capable of assuming the discrete values x_i ($i = 1, 2, \dots, n$). We are not given the corresponding probabilities p_i ; all we know is the expectation value of the function $f(x)$: $\langle f(x) \rangle = \sum p_i f(x_i)$. On the basis of this information, what is the expectation of the function $g(x)$?"

"..In making inference on the basis of partial information we must use the probability distribution which has maximum entropy subject of whatever is known."

Jaynes 1957

<https://bayes.wustl.edu/etj/articles/theory.1.pdf>

Maximum Entropy - Discrete Case

Idea: Find distribution that maximizes the entropy

$$H(X) = - \sum_{i \geq 1} p(x_i) \log p(x_i) .$$

with the following constraints:

- $\sum_{i=1}^n p(x_i) f_k(x_i) = F_k \quad k = 1, \dots, m$
- $\sum_{i=1}^n p(x_i) = 1 .$

Maximum Entropy - Discrete Case Solution

$$p(x_i) = \frac{1}{Z(\lambda_1, \dots, \lambda_m)} \exp(\lambda_1 f_1(x_i) + \dots + \lambda_m f_m(x_i)) . \quad (40)$$

This distribution is called the *Gibbs distribution* in statistical mechanics.

$$Z(\lambda_1, \dots, \lambda_m) = \sum_{i=1}^n \exp(\lambda_1 f_1(x_i) + \dots + \lambda_m f_m(x_i)) , \quad (41)$$

and is called the *partition function*.

$$F_k = \frac{\partial}{\partial \lambda_k} \log Z(\lambda_1, \dots, \lambda_m) . \quad (42)$$

Maximum Entropy - Continuous Case

Idea: Find distribution that maximizes the entropy

$$H(X) = - \int_{-\infty}^{\infty} p(x) \log p(x) dx$$

Instead of the entropy, we use the KL divergence.

- $\int p(x) f_k(x) dx = F_k \quad k = 1, \dots, m .$
- Furthermore, the probability must sum to one:
 $\int p(x) dx = 1 .$

Maximum Entropy - Continuous Case

The probability density function p with maximum D_{KL} subject to these constraints is

$$p(x) = \frac{1}{Z(\lambda_1, \dots, \lambda_m)} m(x) \exp(\lambda_1 f_1(x) + \dots + \lambda_m f_m(x)) \quad (43)$$

with the partition function Z given by

$$Z(\lambda_1, \dots, \lambda_m) = \int m(x) \exp(\lambda_1 f_1(x) + \dots + \lambda_m f_m(x)) dx . \quad (44)$$

Motivation for the EM Algorithm

- likelihood can not be computed analytically
 - hidden states
 - non-linearities
 - many-to-one output mapping
- hidden variables, latent variables, unobserved variables $\mathbf{u}$
- joint probability $p(\mathbf{u}, \mathbf{x}; \mathbf{w})$ is easier to compute than the likelihood
- estimate $p(\mathbf{u}|\mathbf{x}; \mathbf{w})$ by $Q(\mathbf{u} | \mathbf{x})$

EM Algorithm

The EM algorithm is an iteration between two steps, the “E”-step and the “M”-step:

E-step: (45)

$$Q_{k+1} = \arg \max_Q \mathcal{F}(Q, \mathbf{w}_k)$$

M-step:

$$\mathbf{w}_{k+1} = \arg \max_{\mathbf{w}} \mathcal{F}(Q_{k+1}, \mathbf{w}) .$$

It is important to note that in the E-step the maximal Q is

$$Q_{k+1}(\mathbf{u} \mid \{\mathbf{x}\}) = p(\mathbf{u} \mid \{\mathbf{x}\}; \mathbf{w}_k) \quad (46)$$

$$\mathcal{F}(Q_{k+1}, \mathbf{w}_k) = \ln \mathcal{L}(\{\mathbf{x}\}; \mathbf{w}_k) .$$

Some more questions to ask

- Why is KL divergence not a metric?
- What must a metric fulfill?
- Relationship between MLE and KL divergence.
- How to show convexity of logistic regression.
- Describe the Laplacian noise model.
- Basics like Convexity, PSD ...

Relation MLE and KL divergence

Data $x_1, \dots, x_N$ is given following some unknown distribution p . We try to find parameters θ for a distribution q from a family of distributions that best explains the observed data:

$$\begin{aligned}\theta_{ML} &= \arg \max_{\theta} \prod_{i=1}^N q(x_i|\theta) \\&= \arg \max_{\theta} \sum_{i=1}^N \log q(x_i|\theta) \\&= \arg \max_{\theta} \frac{1}{N} \left(\sum_{i=1}^N \log q(x_i|\theta) - \sum_{i=1}^N \log p(x_i) \right) \\&= \arg \min_{\theta} \frac{1}{N} \left(\sum_{i=1}^N \log \frac{p(x_i)}{q(x_i|\theta)} \right) \\&\Rightarrow \arg \min_{\theta} \int \log \left(\frac{p(x_i)}{q(x_i|\theta)} \right) p(x) dx = KL(p||q_{\theta})\end{aligned}$$

Handwritten notes:

- multiplying with / subtracting terms independent of θ does not change $\arg \max_{\theta}$
- $\log \left(\frac{q(x_i|\theta)}{p(x_i)} \right)$
- $= -\log \left(\frac{p(x_i)}{q(x_i|\theta)} \right)$
- $\arg \min_{\theta}$
- with $N \rightarrow \infty$ mean converges to expectation

Gaussian Noise

*s is the 'real' data
y is the observed
noisy data*

*Assumption:
 ϵ is normal distributed*

$$s = X w$$

$$y = s + \epsilon = X w + \epsilon,$$

$$\epsilon \sim N(0, \Sigma)$$

- s is the true signal,
- X is the observed data,
- w is the parameter vector,
- y is the observed target

Goal here:

Assumption about noise

$\Rightarrow$ show that $\hat{\theta}_{ML}$ is equivalent to a loss fct

Either:

minimize loss fct or

ML Estimator of noise model

Gaussian Noise II

The likelihood of $(\mathbf{y}, \mathbf{X})$ is

$$\mathcal{L}((\mathbf{y}, \mathbf{X}); \mathbf{w}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{y} - \mathbf{X} \mathbf{w})^T \Sigma^{-1} (\mathbf{y} - \mathbf{X} \mathbf{w}) \right) .$$

The log-likelihood is

$$\begin{aligned} \ln \mathcal{L}((\mathbf{y}, \mathbf{X}); \mathbf{w}) = \\ - \frac{d}{2} \ln (2\pi) - \frac{1}{2} \ln |\Sigma| - \frac{1}{2} (\mathbf{y} - \mathbf{X} \mathbf{w})^T \Sigma^{-1} (\mathbf{y} - \mathbf{X} \mathbf{w}) . \end{aligned}$$

Gaussian Noise III

Minimize

$$\arg \min_w -\ln \mathcal{L}([y, X]; w) =$$

$$= \arg \min_w (y - Xw)^T \Sigma^{-1} (y - Xw)$$

$$= \arg \min_w \underbrace{y^T \Sigma^{-1} y - 2y^T \Sigma^{-1} Xw + w^T X^T \Sigma^{-1} Xw}_{\text{Log Like}}$$

and the derivative with respect to w is

$$\frac{\partial}{\partial w} \text{Log Like} = -2X^T \Sigma^{-1} y + 2X^T \Sigma^{-1} Xw = 0$$

the least square estimator



$$\underbrace{\hat{w} = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} y}_{\text{ML Estimator}}$$

Gaussian Noise

With the same noise assumption for each observation

$$\Sigma^{-1} = \frac{1}{\sigma} I, \quad \Rightarrow \text{errors are independent of each other}$$

we obtain

$$\hat{w} = (X^T X)^{-1} X^T y,$$

where $(X^T X)^{-1} X^T$ is known as the pseudo inverse or Moore-Penrose inverse of X . The minimal value is

$$\frac{1}{\sigma} y^T \left(I - X (X^T X)^{-1} X^T \right) y.$$

Laplacian Loss Function

The loss function

$$\|y - g(X; w)\|_1$$

corresponds to Laplace noise assumption. Or for one dimension

$$|y - g(x; w)| \text{ .}$$

Laplace Distribution

A random variable has a Laplacian distribution $x \sim \text{Laplace}(\mu, b)$ if its probability density function is

$$p(x \mid \mu, b) = \frac{1}{2b} \exp \left(-\frac{|x - \mu|}{b} \right),$$

$$p(x \mid \mu, b) = \frac{1}{2b} \begin{cases} \exp \left(-\frac{\mu - x}{b} \right) & \text{if } x < \mu \\ \exp \left(-\frac{x - \mu}{b} \right) & \text{if } x \geq \mu \end{cases}$$

- μ is a location parameter and $b > 0$, is a scale parameter.
- $\mu = 0$ and $b = 1$ - positive half-line is exactly an exponential distribution scaled by 1/2.
- Mean is μ , Variance is $2b^2$

Check notation

Likelihood

$$\begin{aligned}\mathcal{L}((\mathbf{y}, \mathbf{X}); \mathbf{w}) &= \prod_{i=1}^N \frac{\beta}{2} \exp(-\beta |y_i - g(\mathbf{x}; \mathbf{w})|) \\ &= \left(\frac{\beta}{2}\right)^N \exp\left(-\beta \sum_{i=1}^N |y_i - g(\mathbf{x}; \mathbf{w})|\right)\end{aligned}$$

Negative log Likelihood

$$\sum_{i=1}^N \frac{\beta}{2} \exp(-\beta |y - g(\mathbf{x}; \mathbf{w})|) + N \log(2b)$$

Maximum likelihood can be obtained by minimizing the sum of absolute values of residuals. (Fitting a linear model, that is robust to outliers.)

For one dimensional output we obtain for the Laplacian noise model

$$p(y - g(\mathbf{x}; \mathbf{w})) = \frac{\beta}{2} \exp(-\beta |y - g(\mathbf{x}; \mathbf{w})|)$$

with loss function

$$|y - g(\mathbf{x}; \mathbf{w})| .$$

Minkowski Loss

The generalization of the squared loss is called **Minkowski loss**.

$$|y - g(\mathbf{x}; \mathbf{w})|^r \quad (47)$$

the corresponding noise model is

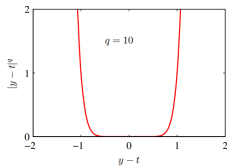
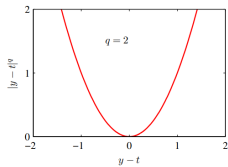
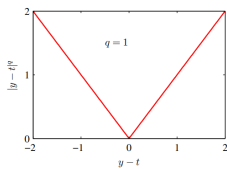
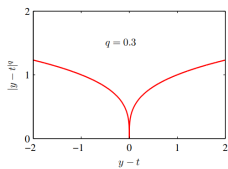
$$p(y - g(\mathbf{x}; \mathbf{w})) = \frac{r \beta^{1/r}}{2 \Gamma(1/r)} \exp(-\beta |y - g(\mathbf{x}; \mathbf{w})|^r) , \quad (48)$$

where Γ is the gamma function

$$\begin{aligned} \Gamma(a) &= \int_0^\infty u^{a-1} e^{-u} du \\ \Gamma(n) &= (n-1)! . \end{aligned} \quad (49)$$

Minkowski Loss

- $r = 2$ quadratic error
- $r < 2$ large errors are down-weighted
- $r > 2$ vice versa
- Which r to use for data with outliers?



(Bishop 1.29)

Cross Entropy Loss

Assuming that the model output is a probability:

$$g_k(\mathbf{x}; \mathbf{w}) = p(\mathbf{y} = \mathbf{e}_k \mid \mathbf{x}) .$$

and that

$$\mathbf{y} \in \{\mathbf{e}_1, \dots, \mathbf{e}_K\} .$$

If $\mathbf{x}$ is in the k -th class then $\mathbf{y} = (0, \dots, 0, 1, 0, \dots, 0)$, where the “1” is at position k in the vector $\mathbf{y}$.

Cross Entropy Loss

The likelihood of iid data is

$$\begin{aligned}\mathcal{L}(\{\mathbf{z}\}; \mathbf{w}) &= p(\{\mathbf{z}\}; \mathbf{w}) = \prod_{i=1}^l \prod_{k=1}^K p(\mathbf{y}^i = \mathbf{e}_k \mid \mathbf{x}^i; \mathbf{w})^{[\mathbf{y}^i]_k} p(\mathbf{x}^i) = \\ &\prod_{i=1}^l p(\mathbf{x}^i) \prod_{k=1}^K p(\mathbf{y}^i = \mathbf{e}_k \mid \mathbf{x}^i; \mathbf{w})^{[\mathbf{y}^i]_k}\end{aligned}$$

The log-likelihood is

$$\ln \mathcal{L}(\{\mathbf{z}\}; \mathbf{w}) = \sum_{k=1}^K \sum_{i=1}^l [\mathbf{y}^i]_k \ln p(\mathbf{y}^i = \mathbf{e}_k \mid \mathbf{x}^i; \mathbf{w}) + \sum_{i=1}^l \ln p(\mathbf{x}^i) .$$

Cross Entropy Loss

Therefore the criterion

$$-\sum_{k=1}^K \sum_{i=1}^l [\mathbf{y}^i]_k \ln p(\mathbf{y}^i = \mathbf{e}_k \mid \mathbf{x}^i; \mathbf{w})$$

is a natural loss function which is called *cross entropy*.

Cross-Entropy - Definition

The *cross entropy* between two distributions p_1 and p_2 is defined as

$$H^*(p_1, p_2) = - \int_U p_1(\mathbf{u}) \ln p_2(\mathbf{u}) d\mathbf{u} .$$

Logistic Regression I

A function g mapping x onto $\mathbb{R}$ can be transformed into a probability by the sigmoidal function

$$p(y = 1 \mid x; w) = \frac{1}{1 + e^{-g(x;w)}}$$

and

$$p(y = 0 \mid x; w) = \frac{e^{-g(x;w)}}{1 + e^{-g(x;w)}}.$$

→ ensures that sums up to 1

Logistic Regression II

We obtain the log odd

$$g(\mathbf{x}; \mathbf{w}) = \ln \left(\frac{p(y = 1 \mid \mathbf{x})}{1 - p(y = 1 \mid \mathbf{x})} \right) .$$

Logistic Regression II

The log-likelihood is

$$\begin{aligned}\ln \mathcal{L}(\{z\}; \mathbf{w}) &= \sum_{i=1}^l \ln p(\mathbf{z}_i; \mathbf{w}) = \sum_{i=1}^l \ln p(y^i, \mathbf{x}^i; \mathbf{w}) = \\ &\sum_{i=1}^l \ln p(y^i | \mathbf{x}^i; \mathbf{w}) + \sum_{i=1}^l \ln p(\mathbf{x}^i) .\end{aligned}$$

Therefore maximum likelihood maximizes

$$\sum_{i=1}^l \ln p(y^i | \mathbf{x}^i; \mathbf{w})$$

*only terms
depending on
 $\mathbf{w}$ are important
for maximizing*

Logistic Regression III

$$\sum_{i=1}^l \ln p(y^i \mid \mathbf{x}^i; \mathbf{w}) = \tag{50}$$
$$\sum_{i=1}^l y^i \ln p(y = 1 \mid \mathbf{x}^i; \mathbf{w}) + \sum_{i=1}^l (1 - y^i) \ln p(y = 0 \mid \mathbf{x}^i; \mathbf{w})$$

Logistic Regression IV

Which gives for the derivative (details can be found in the notes)

$$\frac{\partial}{\partial w_j} \sum_{i=1}^l \ln p(y^i | \mathbf{x}^i; \mathbf{w}) = \sum_{i=1}^l (p(y = 1 | \mathbf{x}^i; \mathbf{w}) - y^i) \frac{\partial g(\mathbf{x}^i; \mathbf{w})}{\partial w_j},$$

where

$$p(y = 1 | \mathbf{x}^i; \mathbf{w}) = \frac{1}{1 + e^{-g(\mathbf{x}^i; \mathbf{w})}}.$$

Logistic Regression

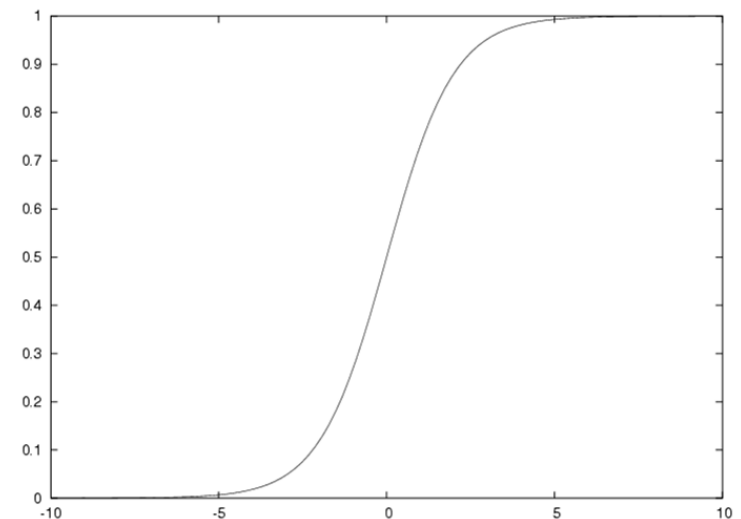
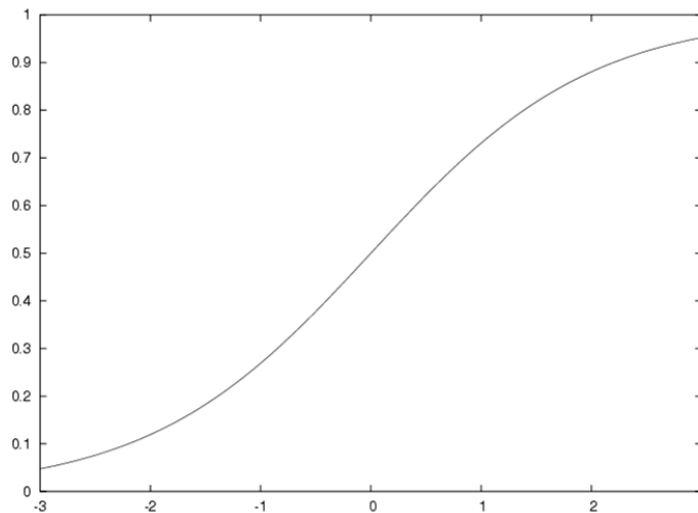
- 4 Noise Models
 - 4.1 Gaussian Noise
 - 4.2 Laplace Noise
 - 4.3 Binary Models
 - 4.3.1 Cross-Entropy
 - 4.3.2 Logistic Regression
 - 4.3.3 Log. Regression Convex
 - 4.3.4 Softmax
 - 4.3.5 Softmax Convex

- 5 Statistical Learning Theory
 - 5.1 Error Bound Example
 - 5.2 Empirical Risk Minimization
 - 5.2.1 Complexity: Finite Number of Functions
 - 5.2.2 Complexity: VC-Dimension
 - 5.3 Error Bounds
 - 5.4 Structural Risk Minimization
 - 5.5 Margin
 - 5.6 Average Bounds

- 6 Kernels and Dot Products
 - 6.1 Mercer's Theorem
 - 6.2 Reproducing Kernel Hilbert Space

a function g mapping x onto $\mathbb{R}$ can be transformed into a probability:

$$p(y = 1 \mid x; w) = \frac{1}{1 + e^{-g(x; w)}}$$



Logistic Regression



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If follows:

$$g(\mathbf{x}; \mathbf{w}) = \ln \left(\frac{p(y = 1 | \mathbf{x})}{1 - p(y = 1 | \mathbf{x})} \right) .$$

log-likelihood:

$$\ln \mathcal{L}(\{\mathbf{z}\}; \mathbf{w}) = \sum_{i=1}^l \ln p(\mathbf{z}_i; \mathbf{w}) = \sum_{i=1}^l \ln p(y^i, \mathbf{x}^i; \mathbf{w}) =$$
$$\sum_{i=1}^l \ln p(y^i | \mathbf{x}^i; \mathbf{w}) + \sum_{i=1}^l \ln p(\mathbf{x}^i)$$

maximum likelihood maximizes

$$\sum_{i=1}^l \ln p(y^i | \mathbf{x}^i; \mathbf{w})$$

Logistic Regression

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derivative of the log-likelihood:

$$\frac{\partial}{\partial w_j} \sum_{i=1}^l \ln p(y^i | \mathbf{x}^i; \mathbf{w}) = \sum_{i=1}^l (p(y = 1 | \mathbf{x}^i; \mathbf{w}) - y^i) \frac{\partial g(\mathbf{x}^i; \mathbf{w})}{\partial w_j}$$

↳ no exam question

similar to the derivative of the quadratic loss function in the regression:

$$\frac{1}{2} \frac{\partial}{\partial w_j} \sum_{i=1}^l (g(\mathbf{x}^i; \mathbf{w}) - y^i)^2 = \sum_{i=1}^l (g(\mathbf{x}^i; \mathbf{w}) - y^i) \frac{\partial g(\mathbf{x}^i; \mathbf{w})}{\partial w_j}$$

$(g(\mathbf{x}^i; \mathbf{w}) - y^i)$ instead of $(p(y = 1 | \mathbf{x}^i; \mathbf{w}) - y^i)$

$$p(y = 1 | \mathbf{x}; \mathbf{w}) = \frac{1}{1 + e^{-g(\mathbf{x}; \mathbf{w})}}$$

Logistic Regression

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6 Kernels and Dot Products

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linear case: $g(\mathbf{x}^i; \mathbf{w}) = \mathbf{w}^T \mathbf{x}^i$

labels: $y \in \{+1, -1\}$

$$\frac{\partial L}{\partial w_j} = - \sum_{i=1}^l y^i x_{ij} (1 - p(y^i | \mathbf{x}^i; \mathbf{w})) \quad (\text{We drop } i \text{ from } \mathbf{x} !)$$

second order derivatives with $H_{jk} = \frac{\partial L}{\partial w_j} \frac{\partial L}{\partial w_k} = \sum_{i=1}^l (y^i)^2 x_{ij} x_{ik} p(y^i | \mathbf{x}; \mathbf{w}) (1 - p(y^i | \mathbf{x}; \mathbf{w}))$
 H as the Hessian

$$\rho_{ij} = x_{ij} \sqrt{p(y^i | \mathbf{x}; \mathbf{w}) (1 - p(y^i | \mathbf{x}; \mathbf{w}))}$$

bilinear form of the Hessian with a vector: $\rightarrow$ concept is important, not details

$$\mathbf{a}^T \mathbf{H} \mathbf{a} = \sum_{i=1}^l \sum_{j=1}^d \sum_{k=1}^d x_{ij} x_{ik} a_j a_k p(y^i | \mathbf{x}; \mathbf{w}) (1 - p(y^i | \mathbf{x}; \mathbf{w})) =$$

$$\sum_{i=1}^l \sum_{j=1}^d a_j x_{ij} \sqrt{p(y^i | \mathbf{x}; \mathbf{w}) (1 - p(y^i | \mathbf{x}; \mathbf{w}))} \sum_{k=1}^d a_k x_{ik} \sqrt{p(y^i | \mathbf{x}; \mathbf{w}) (1 - p(y^i | \mathbf{x}; \mathbf{w}))} =$$

$$\sum_{i=1}^l (\mathbf{a}^T \boldsymbol{\rho}_i) (\mathbf{a}^T \boldsymbol{\rho}_i) = \sum_{i=1}^l (\mathbf{a}^T \boldsymbol{\rho}_i)^2 \geq 0$$

$\rightarrow$ Hessian is positive definite

Convexity of logistic regression

Goal:

- We can show that linear logistic regression is strictly convex.
- We can show that linear softmax is strictly convex.
- More details can be found here
<http://www.qwone.com/~jason/writing/>

Basics needed:

- Convexity
- Positive semi definite (PSD)

Convexity

- S is a **convex set** if, for any $x, x' \in S$, we have

$$\lambda x + (1 - \lambda)x' \in S, \forall \lambda \in [0, 1].$$

- A function $f(x)$ is called **convex** if it is defined on a convex set and if, for any $x, y \in S$, and for any $0 \leq \lambda \leq 1$, we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

- A function $f(x)$ is called **strictly convex** if it is defined on a convex set and if, for any $x, y \in S$, and for any $0 \leq \lambda \leq 1$,

$$f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y).$$

Convexity and Hessian

Theorem: Suppose $f : R^n \rightarrow R$ is twice differentiable over its domain. Then f is convex iff $H = \nabla^2 f(\mathbf{x})$ is positive semi-definite for all $\mathbf{x} \in \text{dom}(f)$. Furthermore, f is strictly convex if H is positive definite.

Definite

A symmetric $n \times n$ matrix A is (strictly) positive definite if

$$\mathbf{v}^T \mathbf{A} \mathbf{v} > 0 \quad \forall \quad \mathbf{v} \neq 0.$$

A symmetric $n \times n$ matrix A is **positive semi definite (PSD)** if

$$\mathbf{v}^T \mathbf{A} \mathbf{v} \geq 0 \quad \forall \quad \mathbf{v} \neq 0.$$

A symmetric $n \times n$ matrix A is (strictly) negative definite if

$$\mathbf{v}^T \mathbf{A} \mathbf{v} < 0 \quad \forall \quad \mathbf{v} \neq 0.$$

A symmetric $n \times n$ matrix A is negative-semi definite if

$$\mathbf{v}^T \mathbf{A} \mathbf{v} \leq 0 \quad \forall \quad \mathbf{v} \neq 0.$$

Frame Title

Theorem 2.6.1

"If the function f has a second derivative that is non- negative (positive) over an interval, the function is convex (strictly convex) over that interval."

(Proof just for the interested students: see Cover and Thomas,1991 or 2006)

↳ also discussed the concept of proving
convexity for softmax

important

↳ know formulas but no need to know
all the index shifting ect

Why we use Bayes

Bayes techniques allow

- to introduce a probabilistic framework
- to deal with hyper-parameters
- to supply error bars and confidence intervals for the model output
- to compare different models
- to select relevant features

Notation

- training data $\{z^1, \dots, z^l\}$ with $(z^i = (x^i, y^i))$,
- the matrix of feature vectors $X = (x^1, \dots, x^l)^T$,
- the vector of labels $y = (y^1, \dots, y^l)^T$,
- training data matrix $Z = (z^1, \dots, z^l)$
- Likelihood $\mathcal{L}$

$$\mathcal{L}(\{z\}; w) = p(\{z\}; w) ,$$

- for iid data sampling the likelihood is

$$\mathcal{L}(\{z\}; w) = p(\{z\}; w) = \prod_{i=1}^l p(z^i; w) .$$

Supervised Learning Setting

$$p(\mathbf{z}; \mathbf{w}) = p(\mathbf{x}) p(y | \mathbf{x}; \mathbf{w}) \quad (51)$$

and

$$\mathcal{L}(\{\mathbf{z}\}; \mathbf{w}) = \prod_{i=1}^l p(\mathbf{x}^i) \prod_{i=1}^l p(y^i | \mathbf{x}^i; \mathbf{w}) . \quad (52)$$

As $\prod_{i=1}^l p(\mathbf{x}^i)$ is independent of the parameters, it is sufficient to maximize the conditional likelihood

$$\mathcal{L}(\{y\} | \{\mathbf{x}\}; \mathbf{w}) = \prod_{i=1}^l p(y^i | \mathbf{x}^i; \mathbf{w}) . \quad (53)$$

Likelihood, Prior, Posterior

$$p(\mathbf{w}|\{\mathbf{z}\}) = \frac{p(\{\mathbf{z}\}|\mathbf{w}) p(\mathbf{w})}{p_{\mathbf{w}}(\{\mathbf{z}\})}$$

$$posterior = \frac{likelihood \times prior}{evidence}$$

- posterior distribution $p(\mathbf{w}|\{\mathbf{z}\})$
- prior distribution $p(\mathbf{w})$
- likelihood $p(\{\mathbf{z}\}|\mathbf{w})$
- evidence or normalization factor

$$\int_W p(\{\mathbf{z}\}|\mathbf{w}) p(\mathbf{w}) d\mathbf{w}$$

Prior Distributions

- Conjugate Prior - posterior is computable in closed form.
(posterior and prior are in the same parameterized family)
- Informative Prior
- Non-informative (Flat) Prior - equal probability to all values
- Improper Priors - can not be normalized

Example Conjugate Prior

- https://en.wikipedia.org/wiki/Conjugate_prior
- What is the conjugate prior for a Poisson likelihood?

Maximum A Posteriori Approach I

Given a dataset $\{\mathbf{x}\} = (x_1, x_2, \dots, x_N)$ of N independent observations.

- ML estimate: choose $\mathbf{w}$ such that data likelihood is maximized:

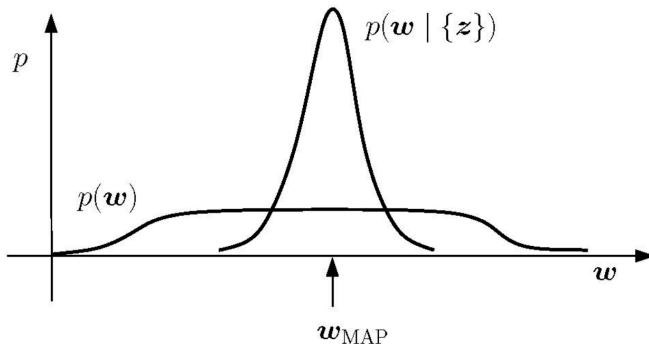
$$\mathbf{w}_{ML} = \arg \max \mathcal{L}(\{\mathbf{x}\}; \mathbf{w})$$

- MAP estimate: choose most probable $\mathbf{w}$ given the data.

$$\mathbf{w}_{MAP} = \arg \max \mathcal{L}(\mathbf{w}; \{\mathbf{x}\})$$

Maximum A Posteriori Approach II

Maximum A Posteriori Approach (MAP):
maximal posterior $p(\mathbf{w} \mid \{\mathbf{z}\})$



MAP III - Regularization

Weight decay (Ridge Regression) - to prevent overfitting

$$\begin{aligned}\tilde{R} &= R(\mathbf{w}) + \lambda\Omega \\ \Omega(\mathbf{w}) &= \|\mathbf{w}\|^2 = \mathbf{w}^T \mathbf{w} = \sum_{ij} w_{ij}^2 .\end{aligned}$$

Maximum A Posteriori Approach IV

Gaussian weight prior

$$p(\mathbf{w}) = \frac{1}{Z_w(\alpha)} \exp\left(-\frac{1}{2} \alpha \|\mathbf{w}\|^2\right) \quad (54)$$

$$Z_w(\alpha) = \int_W \exp\left(-\frac{1}{2} \alpha \|\mathbf{w}\|^2\right) d\mathbf{w} = \left(\frac{2\pi}{\alpha}\right)^{W/2}.$$

- α is a hyper-parameter
- large $\|\mathbf{w}\|$ lead to small $p(\mathbf{w})$ -> we expect smaller weights

Maximum A Posteriori Approach V

Other weight decay terms: Laplace distribution ($\Omega(\mathbf{w}) = \|\mathbf{w}\|_1$)

$$\begin{aligned} p(\mathbf{w}) &= \frac{1}{Z_w(\alpha)} \exp\left(-\frac{1}{2} \alpha \|\mathbf{w}\|_1\right) \\ Z_w(\alpha) &= \int_W \exp\left(-\frac{1}{2} \alpha \|\mathbf{w}\|_1\right) d\mathbf{w} \end{aligned} \quad (55)$$

Cauchy distribution ($\Omega(\mathbf{w}) = \log(1 + \|\mathbf{w}\|^2)$):

$$\begin{aligned} p(\mathbf{w}) &= \frac{1}{Z_w(\alpha)} \exp\left(-\frac{1}{2} \alpha\right) (1 + \|\mathbf{w}\|^2) \\ Z_w(\alpha) &= \int_W \exp\left(-\frac{1}{2} \alpha\right) (1 + \|\mathbf{w}\|^2) d\mathbf{w} . \end{aligned} \quad (56)$$

Maximum A Posteriori Approach VI

Gaussian noise model

$$p(\{z\} \mid \mathbf{w}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (\mathbf{y} - \mathbf{X} \mathbf{w})^T \Sigma^{-1} (\mathbf{y} - \mathbf{X} \mathbf{w})\right) p(\{x\}) \quad (57)$$

and for $\Sigma = \sigma^2 \mathbf{I}$

$$p(\{z\} \mid \mathbf{w}) = \frac{1}{(2\pi)^{d/2} \sigma^d} \exp\left(-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X} \mathbf{w})^T (\mathbf{y} - \mathbf{X} \mathbf{w})\right) p(\{x\}) . \quad (58)$$

The term $R_{\text{emp}} = (\mathbf{y} - \mathbf{X} \mathbf{w})^T (\mathbf{y} - \mathbf{X} \mathbf{w})$ is only the mean squared error.

Maximum A Posteriori Approach VII

- The negative log-posterior is

$$-\log p(\mathbf{w} \mid \{\mathbf{z}\}) = -\log p(\{\mathbf{z}\} \mid \mathbf{w}) - \log p(\mathbf{w}) + \log p_{\mathbf{w}}(\{\mathbf{z}\})$$

- $p_{\mathbf{w}}(\{\mathbf{z}\})$ does not depend on $\mathbf{w}$.
- Minimizing empirical error plus complexity term

$$\tilde{R}(\mathbf{w}) = \frac{1}{2\sigma^2} R_{\text{emp}} + \frac{1}{2} \alpha \Omega(\mathbf{w}) \quad (59)$$

$$= \frac{1}{2} \beta R_{\text{emp}} + \frac{1}{2} \alpha \Omega(\mathbf{w}), \quad (60)$$

where $\beta^{-1} = \sigma^2$.

- Setting $R(\mathbf{w}) = \tilde{R}(\mathbf{w}) 2\sigma^2$ and $\lambda = \sigma^2 \alpha$ we have to minimize

$$R(\mathbf{w}) = R_{\text{emp}} + \lambda \Omega(\mathbf{w}). \quad (61)$$

MAP VIII

Likelihood is the exponential function with empirical error as argument

$$p(\{z\} \mid \mathbf{w}) = \frac{1}{Z_R(\beta)} \exp \left(- \frac{1}{2} \beta R_{\text{emp}} \right) \quad (62)$$

and the prior is an exponential function of the complexity

$$p(\mathbf{w}) = \frac{1}{Z_w(\alpha)} \exp \left(- \frac{1}{2} \alpha \Omega(\mathbf{w}) \right) \quad (63)$$

and the posterior is

$$p(\mathbf{w} \mid \{z\}) = \frac{1}{Z(\alpha, \beta)} \exp \left(- \frac{1}{2} (\alpha \Omega(\mathbf{w}) + \beta R_{\text{emp}}) \right) , \quad (64)$$

where

$$Z(\alpha, \beta) = \int_{\mathcal{W}} \exp \left(- \frac{1}{2} (\alpha \Omega(\mathbf{w}) + \beta R_{\text{emp}}) \right) d\mathbf{w} . \quad (65)$$

Posterior Approximation

In order to approximate the posterior a Gaussian assumption is made.

First we make a Taylor expansion of $\tilde{R}(\boldsymbol{w})$ around its minimum $\boldsymbol{w}_{\text{MAP}}$:

$$\tilde{R}(\boldsymbol{w}) = \tilde{R}(\boldsymbol{w}_{\text{MAP}}) + \frac{1}{2} (\boldsymbol{w} - \boldsymbol{w}_{\text{MAP}})^T \boldsymbol{H} (\boldsymbol{w} - \boldsymbol{w}_{\text{MAP}}) , \quad (66)$$

where the first order derivatives vanish at the minimum and $\boldsymbol{H}$ is the Hessian of $\tilde{R}(\boldsymbol{w})$ at $\boldsymbol{w}_{\text{MAP}}$.

Posterior Approximation II

The posterior is now a Gaussian

$$p(\mathbf{w} \mid \{\mathbf{z}\}) = \frac{1}{Z} \exp(-\tilde{R}(\mathbf{w})) = \tag{67}$$
$$\frac{1}{Z} \exp\left(-\tilde{R}(\mathbf{w}_{\text{MAP}}) - \frac{1}{2} (\mathbf{w} - \mathbf{w}_{\text{MAP}})^T \mathbf{H} (\mathbf{w} - \mathbf{w}_{\text{MAP}})\right),$$

where Z is a normalization constant.

The Hessian for weight decay is given by

$$\mathbf{H} = \frac{1}{2\sigma^2} \mathbf{H}_{\text{emp}} + \alpha \mathbf{I} = \frac{\beta}{2} \mathbf{H}_{\text{emp}} + \alpha \mathbf{I}, \tag{68}$$

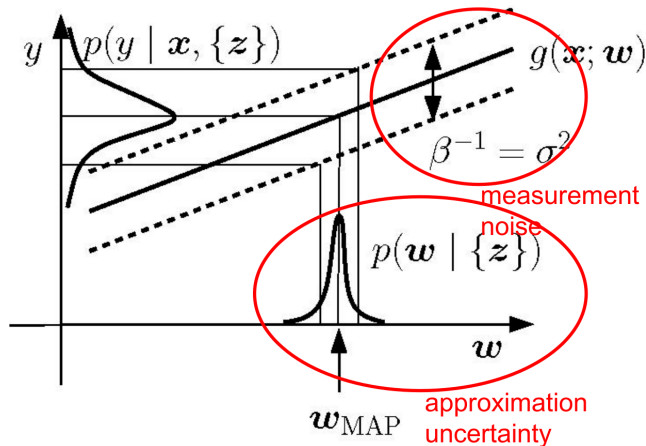
where $\mathbf{H}_{\text{emp}}$ is the Hessian of the empirical error.

Posterior Approximation III

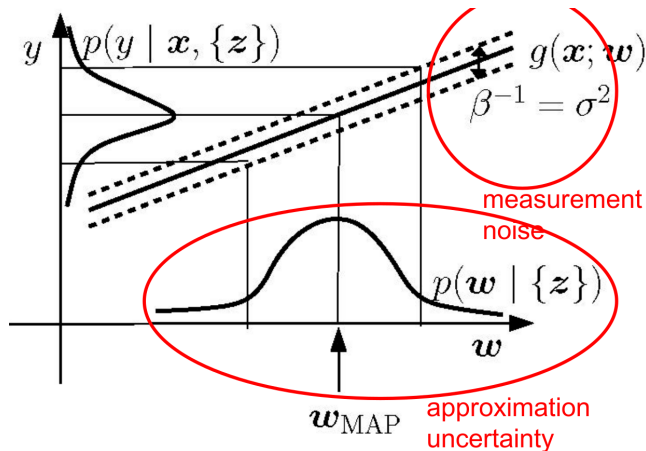
The normalization constant is

$$Z(\alpha, \beta) = \exp \left(-\tilde{R}(\mathbf{w}_{\text{MAP}}) \right) (2\pi)^{W/2} |\mathbf{H}|^{-1/2} . \quad (69)$$

Error bars and confidence intervals I



Error bars and confidence intervals II



Evidence Framework I

- Hyperparameters α and β are unknown
- β is an assumption on the noise in the data.
- α is an assumption on the optimal network complexity relative to the empirical error.

$$\begin{aligned} p(\mathbf{w} \mid \{\mathbf{z}\}) &= \\ \int_{S_\alpha} \int_{S_\beta} p(\mathbf{w}, \alpha, \beta \mid \{\mathbf{z}\}) d\alpha d\beta &= \\ \int_{S_\alpha} \int_{S_\beta} p(\mathbf{w} \mid \alpha, \beta, \{\mathbf{z}\}) p(\alpha, \beta \mid \{\mathbf{z}\}) d\alpha d\beta . \end{aligned} \tag{70}$$

Evidence Framework II

- Approximation of the posterior
- Integration over hyperparameters

EF III - Approximation of the posterior

- Assume sharply peaked values around α_{MAP} and β_{MAP}

$$\begin{aligned} p(\mathbf{w} \mid \{\mathbf{z}\}) &\cong p(\mathbf{w} \mid \alpha_{\text{MAP}}, \beta_{\text{MAP}}, \{\mathbf{z}\}) \int_{S_\alpha} \int_{S_\beta} p(\alpha, \beta \mid \{\mathbf{z}\}) d\alpha d\beta \\ &= p(\mathbf{w} \mid \alpha_{\text{MAP}}, \beta_{\text{MAP}}, \{\mathbf{z}\}) . \end{aligned}$$

EF III - Approximation of Posterior II

The posterior of α and β is

$$p(\alpha, \beta \mid \{\mathbf{z}\}) = \frac{p(\{\mathbf{z}\} \mid \alpha, \beta) p(\alpha, \beta)}{p_{\alpha, \beta}(\{\mathbf{z}\})} . \quad (71)$$

We express the (α, β) -likelihood through marginalization over $\mathbf{w}$:

$$\begin{aligned} p(\{\mathbf{z}\} \mid \alpha, \beta) &= \int_W p(\{\mathbf{z}\} \mid \mathbf{w}, \alpha, \beta) p(\mathbf{w} \mid \alpha, \beta) d\mathbf{w} = \quad (72) \\ &\int_W p(\{\mathbf{z}\} \mid \mathbf{w}, \beta) p(\mathbf{w} \mid \alpha) d\mathbf{w} . \end{aligned}$$

EF III - Approximation of Posterior III

We obtain

$$p(\{\mathbf{z}\} \mid \alpha, \beta) = \frac{Z(\alpha, \beta)}{Z_w(\alpha) Z_R(\beta)} . \quad (73)$$

Especially,

$$p(\mathbf{w} \mid \{\mathbf{z}\}) = \frac{Z(\alpha, \beta)}{Z_R(\beta) Z_w(\alpha)} p(\{\mathbf{z}\} \mid \mathbf{w}) p(\mathbf{w}) . \quad (74)$$

Example

We will show an example with concrete empirical error and prior term. See lecture notes.

EF III - Integration over hyperparameters

$$\begin{aligned} p(\mathbf{w} \mid \{\mathbf{z}\}) &= \int_{S_\alpha} \int_{S_\beta} p(\mathbf{w}, \alpha, \beta \mid \{\mathbf{z}\}) d\alpha d\beta = \\ &\int_{S_\alpha} \int_{S_\beta} p(\mathbf{w} \mid \alpha, \beta, \{\mathbf{z}\}) p(\alpha, \beta \mid \{\mathbf{z}\}) d\alpha d\beta = \\ &\frac{1}{p_w(\{\mathbf{z}\})} \int_{S_\alpha} \int_{S_\beta} p(\{\mathbf{z}\} \mid \mathbf{w}, \beta) p(\mathbf{w} \mid \alpha) p(\alpha) p(\beta) d\alpha d\beta . \end{aligned}$$

- α and β are independent and do not depend on the data:

$$p(\alpha, \beta \mid \{\mathbf{z}\}) = p(\alpha) p(\beta)$$

- likelihood term is independent of α
- prior of weights is independent of β

Integrating Out - Hyperpriors

- we choose a non-informative prior on a logarithmic scale
- $p(\ln(\alpha))$ and $p(\ln(\beta))$ are constant

$$p(\alpha) = \frac{1}{\alpha}$$
$$p(\beta) = \frac{1}{\beta}$$

EF IV - Integrating Out

We first consider the prior over the weights

$$\begin{aligned} p(\mathbf{w}) &= \int_0^\infty p(\mathbf{w} \mid \alpha) p(\alpha) d\alpha = \\ &\int_0^\infty \frac{1}{Z_w(\alpha)} \exp\left(-\frac{1}{2} \alpha \Omega(\mathbf{w})\right) \frac{1}{\alpha} d\alpha = \\ (2\pi)^{-W/2} \int_0^\infty \exp\left(-\frac{1}{2} \alpha \Omega(\mathbf{w})\right) \alpha^{W/2-1} d\alpha &= \frac{\Gamma(W/2)}{(\pi \Omega(\mathbf{w}))^{W/2}}, \end{aligned} \tag{75}$$

where Γ is the gamma function.

Analog we obtain

$$p(\{\mathbf{z}\} \mid \mathbf{w}) = \frac{\Gamma(l/2)}{(\pi R_{\text{emp}})^{l/2}}. \tag{76}$$

EF IV - Integrating Out II

- Now we combine:

$$p(\mathbf{w}) = \frac{\Gamma(W/2)}{(\pi \Omega(\mathbf{w}))^{W/2}}$$
$$p(\{\mathbf{z}\} \mid \mathbf{w}) = \frac{\Gamma(l/2)}{(\pi R_{\text{emp}})^{l/2}}$$

- which leads to the following:

$$-\ln p(\mathbf{w} \mid \{\mathbf{z}\}) = \frac{l}{2} \ln R_{\text{emp}} + \frac{W}{2} \ln \Omega(\mathbf{w}) + \text{const}$$

- Gradient is given as

$$-\nabla_{\mathbf{w}} \ln p(\mathbf{w} \mid \{\mathbf{z}\}) = \frac{l}{2 R_{\text{emp}}} \nabla_{\mathbf{w}} R_{\text{emp}} + \frac{W}{2 \Omega(\mathbf{w})} \nabla_{\mathbf{w}} \Omega(\mathbf{w}) .$$

EF IV - Integrating Out III

We compare this with

$$- \ln p(\mathbf{w} \mid \{\mathbf{z}\}) = \frac{1}{2} \beta R_{\text{emp}} + \frac{1}{2} \alpha \Omega(\mathbf{w}) .$$

and the gradient

$$- \nabla_{\mathbf{w}} \ln p(\mathbf{w} \mid \{\mathbf{z}\}) = \frac{1}{2} \beta \nabla_{\mathbf{w}} R_{\text{emp}} + \frac{1}{2} \alpha \nabla_{\mathbf{w}} \Omega(\mathbf{w}) .$$

Solving for α and β leads to the *effective* values:

$$\begin{aligned} \alpha^{\text{eff}} &= \frac{W}{\Omega(\mathbf{w}_{\text{MAP}})} \\ \beta^{\text{eff}} &= \frac{l}{R_{\text{emp}}(\mathbf{w}_{\text{MAP}})} . \end{aligned} \tag{77}$$

EF IV - Integrating Out III

Idea: use the actual values of α β to find $\mathbf{W}_{MAP}$ through gradient descent and then update them

$$\alpha^{\text{eff}} = \frac{W}{\Omega(\mathbf{w}_{MAP})}$$
$$\beta^{\text{eff}} = \frac{l}{R_{\text{emp}}(\mathbf{w}_{MAP})} .$$

Model Comparison I

We can compare model classes $\mathcal{M}$.

$$p(\mathcal{M} \mid \{z\}) = \frac{p(\{z\} \mid \mathcal{M}) p(\mathcal{M})}{p_{\mathcal{M}}(\{z\})} ,$$

where $p(\{z\} \mid \mathcal{M})$ is here the likelihood of the data given a model class but is at the same time the evidence

$$p(\{z\} \mid \mathcal{M}) = \int_W p(\{z\} \mid \boldsymbol{w}, \mathcal{M}) p(\boldsymbol{w} \mid \mathcal{M}) d\boldsymbol{w} ,$$

where we only made all probabilities conditioned on the model class $\mathcal{M}$.

Model Comparison II

It can be shown that

$$\begin{aligned} \ln p(\{\mathbf{z}\} \mid \mathcal{M}) = & -\frac{\alpha_{\text{MAP}}}{2} \Omega(\mathbf{w}_{\text{MAP}}) - \frac{\beta_{\text{MAP}}}{2} R_{\text{emp}} - \frac{1}{2} \ln |\mathbf{H}| + \\ & \frac{W}{2} \ln \alpha_{\text{MAP}} + \frac{l}{2} \ln \beta_{\text{MAP}} + \ln M! + 2 \ln M + \frac{1}{2} \ln \left(\frac{2}{\gamma} \right) + \\ & \frac{1}{2} \ln \left(\frac{2}{l - \gamma} \right) . \end{aligned}$$

with M being the number of hidden units in the network

Weight Space Symmetries

- Consider 2 layer network
- Changing the sign of weights
- Input-Output Mapping unchanged
- M hidden units $\rightarrow 2^M$ equivalent weight vectors
- interchanging weights $\rightarrow M!$
- Overall weight space symmetrie $M!2^M$
- Paper: Chen et al. 1993, On the Geometry of Feedforward Neural Network Error Surfaces
(<https://ieeexplore.ieee.org/document/6796044>)

Posterior Sampling I

The question is how to approximate integrals like

$$A(f) = \int_W f(\mathbf{w}) p(\mathbf{w} \mid \{z\}) d\mathbf{w} \quad (78)$$

we can sample weight vectors $\mathbf{w}_i$ to estimate

$$A(f) \approx \frac{1}{L} \sum_{i=1}^L f(\mathbf{w}_i) , \quad (79)$$

where the $\mathbf{w}_i$ are sampled according to $p(\mathbf{w} \mid \{z\})$.

Posterior Sampling I

- Metropolis Algorithm

<https://bayes.wustl.edu/Manual/EquationOfState.pdf>

- Monte Carlo Methods

- Importance Sampling

- Markov Chain Monte Carlo Methods